

Uniform Bi-Hölder Transport from Weak MTW

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Abstract

We prove that the weak Ma–Trudinger–Wang condition on a fixed smooth connected compact boundaryless Riemannian manifold of dimension at least two implies a common Hölder estimate for optimal transport maps and their inverses over the entire class of probability densities with fixed positive upper and lower bounds. The maps have homeomorphic representatives, conjugate cut points are allowed, and no density regularity is assumed.

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1 Introduction

For the squared-distance cost on a compact Riemannian manifold, positive upper and lower density bounds give an almost-everywhere optimal map, but continuity depends on the geometry of the cost. The weak Ma–Trudinger–Wang condition controls that geometry where the cost is smooth. We prove that it gives a common Hölder estimate for the map and its inverse throughout each fixed density-bounded class, even when transport graphs approach conjugate pairs.

Let (M, g) be a smooth connected compact Riemannian manifold without boundary, of dimension $n \geq 2$. Write d and vol for its distance and volume measure, and set

$$c(x, y) = \frac{1}{2}d(x, y)^2, \quad I(x) = \{p \in T_x M : d(x, \exp_x(ap)) = a|p| \text{ for some } a > 1\}.$$

We assume the weak Ma–Trudinger–Wang condition: for every $p \in I(x)$ and $\xi, \eta \in T_x M$ with $\xi \perp \eta$,

$$-\frac{3}{2} \frac{\partial^4}{\partial s^2 \partial t^2} c(\exp_x(t\xi), \exp_x(p + s\eta)) \Big|_{s=t=0} \geq 0. \quad (1.1)$$

The cost is smooth near the pair in this formula. No curvature inequality at a nonsmooth cost pair is assumed. For probability densities ρ_0, ρ_1 , an optimal map is a measurable map T with $T_\#(\rho_0 \text{ vol}) = \rho_1 \text{ vol}$ that minimizes $\int_M c(x, T(x)) \rho_0(x) d\text{vol}(x)$ among maps with that pushforward.

Theorem 1.1 (Uniform bi-Hölder transport). *Fix (M, g) as above and $0 < \lambda \leq \Lambda < \infty$ for which the following density class is nonempty. There are $\alpha \in (0, 1]$ and $C < \infty$, depending only on $(M, g), \lambda, \Lambda$, such that the following holds for every pair of measurable densities satisfying*

$$\int_M \rho_i d\text{vol} = 1, \quad \lambda \leq \rho_i \leq \Lambda \quad \text{vol-almost everywhere}, \quad i = 0, 1.$$

The almost-everywhere unique optimal map from $\rho_0 \text{ vol}$ to $\rho_1 \text{ vol}$ for the cost c has a homeomorphic representative $\tilde{T} : M \rightarrow M$, and

$$d(\tilde{T}(x), \tilde{T}(x')) \leq Cd(x, x')^\alpha, \quad d(\tilde{T}^{-1}(y), \tilde{T}^{-1}(y')) \leq Cd(y, y')^\alpha$$

for all $x, x', y, y' \in M$.

The constants are common to the full density class on the fixed manifold. They do not depend on individual densities or their derivatives. The theorem includes conjugate cut points, and does not assume nonfocality, strict MTW, or cut avoidance. We obtain a positive exponent by contradiction; no explicit exponent or uniformity over varying metrics is asserted.

1.1 Context and geometric input

McCann’s polar factorization theorem supplies the almost-everywhere unique optimal map and its cost potential [12]. The curvature condition originates in the work of Ma–Trudinger–Wang [11]; Loeper established its connection with supporting cost functions and continuity [9]. Sections and contact-volume comparison are central tools in Caffarelli’s theory [1] and in the weak-MTW regularity theory of Figalli–Kim–McCann [5, Theorem 2.1 and Corollary 9.6]. These results already treat rough densities. Their smooth-cost Hölder estimates require the appropriate cost-convexity

and nondegeneracy hypotheses. A density-controlled exponent after localization does not by itself provide a common localization scale or Hölder constant for a family whose transport graphs approach conjugate pairs.

Uniform control over a density-bounded class is known in important geometric settings. Loeper–Villani obtain such estimates under strict MTW and nonfocality, with a further extension under their uniform-regularity hypothesis and positivity of the minimizing-fiber diameter $\delta(M)$ [10, Corollaries 8.3–8.4]. Figalli–Kim–McCann prove uniform cut separation on products of round spheres [6, Theorem 5.1 and Corollary 5.2]. The issue addressed here is the general fixed weak-MTW manifold, without imposing such separation. A different quantitative stability result of Warren [16, Theorem 1] assumes common C^2 norm bounds on the logarithms of smooth densities and sufficiently small Wasserstein distance; those assumptions are not part of the measurable-density class considered here.

The density-free input is proved in the companion paper *Global Support and Convex Injectivity Domains under Weak MTW* [14]. It derives convexity of open and closed injectivity domains, global support for every ordinary subgradient, and uniform intermediate regularity from (1.1). We state the complete interface in Section 2. The global supporting conclusion has an important antecedent: Figalli–Rifford–Villani [7, Lemma 3.1] prove the global double-mountain principle assuming convex injectivity domains as well as weak MTW. The companion supplies the geometric hypothesis under weak MTW alone, together with the intermediate statements needed below. Figalli–Gallouët–Rifford [4, Theorem 1.7] previously proved the convexity implication under nonfocality. On compact surfaces, Figalli–Rifford–Villani [7, Theorem 1.3] proved that transport continuity is equivalent to the conjunction of convex injectivity domains and weak MTW. Their introduction poses the equivalence with weak MTW alone; the separate convexity and continuity questions also appear in [8, Questions 8.6–8.7].

Figalli–Rifford–Villani’s transport continuity property concerns all density pairs bounded above and away from zero. They prove that this property implies convex injectivity domains and weak MTW [7, Definition 1.1 and Theorem 1.2(i)]. Theorem 1.1 implies that property by choosing bounds for each pair. Combined with their necessity result, it yields the weak-MTW–transport-continuity equivalence, with the forward direction strengthened to the class-uniform estimates above.

1.2 Conventions

We use dual potentials with

$$u(x) = \sup_y \{-c(x, y) - v(y)\}, \quad v(y) = \sup_x \{-c(x, y) - u(x)\}.$$

Thus $u(x) + c(x, y) + v(y) \geq 0$, with equality at a contact. The symbol $\partial u(x)$ denotes the ordinary subdifferential of the semiconvex function, with covectors identified with tangent vectors by the metric. At a differentiable contact its gradient is the initial minimizing log. We write

$$G_x = \{p \in T_x M : d(x, \exp_x p) = |p|\} = \overline{I(x)}, \quad c_t = c/t, \quad Q_t f(z) = \min_x \{f(x) + c_t(x, z)\},$$

and, for $p \in I(x)$,

$$A_x(p) = D_{xx}^2 c(x, \exp_x p), \quad \sigma_x(p) = |\det(d \exp_x)_p|.$$

The function σ_x extends continuously to G_x . At an ordinary cut point, a specified nonconjugate minimizing branch has a smooth upper cost branch; we state explicitly when such a branch is used. A star denotes the adjoint. Constants denoted by c, C may change between formulas. Constants in the final comparison depend only on the fixed geometry and density bounds; constants used solely to pass limits at fixed scales may also depend on the already fixed templates.

1.3 Proof structure

For a dual pair (u, v) , the lifted gap section at x and height $r \geq 0$ is

$$\mathcal{S}_x(r) = \left\{ p \in G_x : u(x) + \frac{|p|^2}{2} + v(\exp_x p) \leq r \right\}.$$

We study the oscillation of v on its endpoint image $\exp_x \mathcal{S}_x(r)$. Convexity of the lifted section turns a power bound on that oscillation into a power bound on the diameter of the endpoint image. Taking the oscillation supremum over all admissible density pairs makes this a single scalar estimate for the entire class. Once it is proved, every zero-height section has one endpoint, and comparison of nearby poles gives the Hölder estimate in both transport directions.

The foundation theorem applies to all ordinary subgradients, including convex combinations of all minimizing logs at active cut endpoints. The projection $(x, p) \mapsto \exp_x(tp)$ of the full subgradient graph is a homeomorphism for $0 < t < 1$; only the supporting inequality is asserted at time one for an arbitrary potential. It also gives compact convex lifted gap sections and $C^{1,1}$ intermediate potentials. The minimizing point in $Q_t u(z)$, called its pole, depends Lipschitzly on z .

Section 3 develops the quantitative estimates needed when a minimizing endpoint approaches conjugacy. A fixed joining point factors the full Jacobi matrix into nonsingular piece matrices and a positive semidefinite broken-action Hessian. Radial shortening adds a uniformly positive form. Transverse MTW, the radial identity, and a common separating logarithmic scale then compare ordered singular values along an extended affine segment.

Section 4 takes a supremum of section oscillations over the entire density class. Failure of every power bound gives two separated gap scales on which this supremum is almost constant. Two scalar modifications produce a positive comparison maximum. One modifies the value at a center endpoint; the other supplies the active endpoints whose minimizing logs have the center log as a convex combination. Varying their amplitude gives an inequality for every positive active barycentric representation, with constants fixed before the curvature of the outer modification is chosen.

Section 5 keeps the order of the limiting argument explicit. An exact pole-matrix identity transfers lower jets to active endpoints. The original contact-volume bound first excludes conjugacy at a selected outer endpoint and bounds its determinant. This bounds the center Hessian. Only then does the short-time minimizer have nonsingular derivative, allowing the original almost-everywhere density equation to be used at the sampled center endpoints. An oblique-projection determinant gain and the spectral comparison give the contradiction. Finally, Section 6 converts the resulting power bound into the common estimate for the map and its inverse.

2 The density-free foundation

We record the precise geometric input from [14]. Every statement in this section is density-free and uses only the fixed compact manifold and weak MTW. The proofs, including the treatment of conjugate cut endpoints, are in that paper.

A potential is a function $u(x) = \max_y \{-c(x, y) - w(y)\}$ with a continuous datum w ; replacing w by the cost transform of u gives a dual pair. Put $D_g = \text{diam } M$. The total minimizing-vector set $\{(x, p) : p \in G_x\}$ is compact. For $p \in G_x$ and $y_s = \exp_x(sp)$, $0 < s < 1$, divided action gives

$$c(z, \exp_x p) \leq c_s(z, y_s) + \frac{1-s}{2}|p|^2, \quad c_s(x, y_s) = \frac{s}{2}|p|^2. \quad (2.1)$$

The right side is a smooth upper support near x .

Lemma 2.1 (Uniform supports and active logs). *There is a geometric constant C_g such that the cost is uniformly semiconcave in either variable, including at cut points. Every potential is D_g -Lipschitz and uniformly semiconvex. For a continuous representing datum w , define*

$$\mathcal{V}_u(x) = \{p \in G_x : u(x) + c(x, \exp_x p) + w(\exp_x p) = 0\}.$$

Then $\mathcal{V}_u(x)$ is compact and $\partial u(x) = \text{conv } \mathcal{V}_u(x)$. Every $p \in \partial u(x)$ has a positive representation using at most $n + 1$ affinely independent active logs. The set $\mathcal{V}_u(x)$ includes every minimizing log to each endpoint active for the specified datum w .

This is [14, Lemma 4.1]. In particular, the smooth upper supports obtained by splitting at time $1/2$ have a common neighborhood size and Hessian bound. The representing datum need not itself be a cost potential.

Write $H_s(p) = A_x(sp)/s$. At a specified nonconjugate minimizing branch, A_x will also denote the smooth branch Hessian.

Lemma 2.2 (Radial divided action). *Let $p \in G_x$, and let $S_p(r)$ be the Jacobi tensor along $\exp_x(rp)$ with $S_p(0) = 0$, $D_r S_p(0) = \text{Id}$, in parallel orthonormal frames. For $0 < t < s < 1$,*

$$\frac{d}{dr} H_r(p) = -S_p(r)^{-1} S_p(r)^{-T}, \quad H_t(p) - H_s(p) \geq \kappa_g(s - t)\text{Id},$$

where $\kappa_g > 0$ depends only on the manifold. On every smooth minimizing branch, $A_x(w)w = w$.

Lemma 2.3 (Transverse slack). *Let $p_i \in G_x$, $m_i > 0$, $\sum_i m_i = 1$, and set $p = \sum_i m_i p_i$, $E = \text{span}\{p_i - p\}$. If $t \text{conv}\{p_i\} \subset I(x)$ and $0 < t < s < 1$, then*

$$\left[\frac{A_x(tp)}{t} - \sum_i m_i \frac{A_x(sp_i)}{s} \right] \Big|_{E^\perp} \geq \kappa_g(s - t)\text{Id}.$$

The conclusion also holds at a limiting time t if the radially contracted hulls lie in $I(x)$ and the displayed Hessians have nonconjugate branch limits. No minimizing assumption about sp is needed in this limiting assertion.

These are [14, Lemmas 4.2–4.3]. The latter applies to every positive finite representation, not merely a selected Carathéodory representation. We reconstruct the Jacobi identities and the additional uniform spectral bounds in Section 3.

Lemma 2.4 (Short time for a semiconvex datum). *Let f be Lipschitz and semiconvex. For sufficiently small $t > 0$, depending only on these two bounds and the manifold,*

$$F_t : \Gamma_f \rightarrow M, \quad \Gamma_f = \{(x, p) : p \in \partial f(x)\}, \quad F_t(x, p) = \exp_x(tp)$$

is a homeomorphism with locally Lipschitz inverse into the tangent bundle. The minimizing pole $x_t(z)$ of $Q_t f$ is unique, $Q_t f \in C^{1,1}$, and $\text{grad } Q_t f(z) = -t^{-1} \log_z x_t(z)$. The graph Γ_f is an n -dimensional Lipschitz manifold.

This is [14, Lemma 4.4]. The local mechanism is the curved version of proximal regularization and quadratic inf-convolution; compare [13, Sections 3–5 and 7]. It is needed below for increasing scalar modifications of potentials, which need not themselves be cost potentials.

Theorem 2.5 (Global supporting property). *For every potential u , every $x \in M$, and every $p \in \partial u(x)$, one has $p \in G_x$ and*

$$u(x') \geq u(x) + c(x, \exp_x p) - c(x', \exp_x p) \quad (x' \in M).$$

For each $0 < t < 1$, $F_t : \Gamma_u \rightarrow M$ is a homeomorphism, $tp \in I(x)$, and its poles are the unique global minimizers of $Q_t u$.

Corollary 2.6 (Convex injectivity domains). *Both G_x and $I(x)$ are convex for every $x \in M$.*

Corollary 2.7 (Convex lifted gap sections). *For a dual pair (u, v) , each set*

$$\mathcal{S}_x(r) = \left\{ p \in G_x : u(x) + \frac{|p|^2}{2} + v(\exp_x p) \leq r \right\}$$

is compact and convex, including ordinary and conjugate cut endpoints. If it is nonempty and $r \geq 0$, then

$$\text{diam } \mathcal{S}_x(r)^2 \leq 8 \left(r + \text{osc}_{p \in \mathcal{S}_x(r)} v(\exp_x p) \right).$$

These are [14, Theorem 4.8 and Corollaries 4.9–4.10]. Convexity of injectivity domains alone is not the whole input: the ordinary-subgradient supporting assertion is also used for modified envelopes and local lower tests.

Proposition 2.8 (Intermediate regularity and pole control). *Let (u, v) be a dual pair and $0 < t < 1$. Then $Q_t u = -Q_{1-t} v \in C^{1,1}(M)$, with almost-everywhere bounds*

$$-\frac{C_g}{1-t} g \leq \text{Hess } Q_t u \leq \frac{C_g}{t} g.$$

If $(x_t(z), p_t(z)) = F_t^{-1}(z)$ and Φ_s is geodesic flow on TM , then

$$(x_t(z), p_t(z)) = \Phi_{-t}(z, \text{grad } Q_t u(z)).$$

In particular, the pole and momentum maps are Lipschitz with constants depending only on M, g, t . There are also $r_t, b_t > 0$, depending only on M, g, t , such that every $z = \exp_x(tp)$, $p \in \partial u(x)$, satisfies

$$u(x') + c_t(x', z) \geq u(x) + c_t(x, z) + b_t d(x, x')^2 \quad \text{if } d(x, x') < r_t.$$

All these constants are uniform when t ranges in a compact subinterval of $(0, 1)$.

This is [14, Proposition 4.11]. Its proof uses the two-sided supports from the split contact geodesic and smooth backward geodesic flow, not inversion of the full endpoint exponential map.

Remark 2.9. No time-one injectivity is claimed for an arbitrary potential. For example, $u(x) = -c(x, y_0)$ can have many poles with endpoint y_0 . All inverse projections and $C^{1,1}$ regularizations in this paper are used at times strictly between zero and one.

3 Jacobi fields and uniform comparison of exponential Jacobians

This section proves two estimates that remain uniform near conjugate endpoints: a positive divided-Hessian difference after pullback by an exponential differential, and a comparison of exponential Jacobians along an extended affine segment. The radial estimates use only the fixed compact Riemannian manifold. Weak MTW enters when we compare different rays on the segment. None of

these constants depends on the scalar perturbations introduced in Section 4. The pullback estimate is Lemma 3.4; it supplies the positive term in the center lower bound of Lemma 5.4. The affine-segment estimate is Lemma 3.5. Corollary 3.6 applies it to a positive barycentric representation. A fixed positive vertex weight supplies the opposite extension required for the affine comparison; radial and norm bounds control its remaining hypotheses. This is the geometric comparison needed in the final determinant argument.

For a linear map between tangent spaces, singular values and adjoints are defined using the Riemannian inner products. We write

$$s_1(x, v) \leq \cdots \leq s_n(x, v)$$

for the singular values of $(d \exp_x)_v$, including zero singular values when $v \in G_x$ is conjugate. Thus

$$\sigma_x(v) = \prod_{j=1}^n s_j(x, v).$$

For a positive definite symmetric matrix Q , the notation $\lambda_j^\downarrow(Q)$ lists its eigenvalues in decreasing order. All matrix identities below are written in parallel orthonormal frames; their statements are independent of that choice.

3.1 The divided-action identities

On a minimizing ray, the divided Hessian $S(T)$ below is $H_T(v)$ from Lemma 2.2. We record its Jacobi form together with reversal, and distinguish the local action from the global cost when a branch passes through cut.

Lemma 3.1 (Jacobi matrices and reversal). *Let $\gamma(s) = \exp_x(sv)$ be a geodesic and identify its tangent spaces by parallel transport. Put $\mathcal{R}(s)w = R(w, \dot{\gamma}(s))\dot{\gamma}(s)$ and let C, J solve*

$$U'' + \mathcal{R}U = 0, \quad C(0) = \text{Id}, \quad C'(0) = 0, \quad J(0) = 0, \quad J'(0) = \text{Id}.$$

Then

$$(d \exp_x)_{Tv} = \frac{J(T)}{T} \tag{3.1}$$

for $T > 0$. Whenever the geodesic to $\gamma(T)$ is nonconjugate, let $c_{\gamma, T}$ be its smooth local action: one half the squared length of the geodesic branch obtained by varying the two endpoints. The source Hessian of its divided action is

$$S(T) := D_{xx}^2 \frac{c_{\gamma, T}(x, \gamma(T))}{T} = J(T)^{-1} C(T), \quad S'(T) = -J(T)^{-1} J(T)^{-*}. \tag{3.2}$$

The endpoint is held fixed when taking D_{xx}^2 ; the derivative in T then follows the displayed geodesic. The identities apply to a smooth action branch whether or not it minimizes. Before cut, this action is the cost c ; at a nonconjugate minimizing cut vector, it is the chosen smooth upper branch of c .

If $y = \exp_x v$ and $\bar{v} = -P_{x \rightarrow y} v$ is the reverse initial velocity, then

$$(d \exp_y)_{\bar{v}} = ((d \exp_x)_v)^*, \quad \sigma_y(\bar{v}) = \sigma_x(v), \tag{3.3}$$

with the indicated source and target tangent spaces. Finally, for $v \in I(x)$,

$$A_x(v)v = v. \tag{3.4}$$

Proof. Varying the initial velocity in the geodesic equation gives (3.1). Since $\mathcal{R}$ is symmetric, the fundamental matrix

$$\Phi(T) = \begin{pmatrix} C(T) & J(T) \\ C'(T) & J'(T) \end{pmatrix}$$

is symplectic. In particular,

$$J^* J' = J'^* J, \quad C^* J' - C'^* J = \text{Id}, \quad C J^* = J C'^*. \quad (3.5)$$

These identities can also be obtained by differentiating their left sides and using the Jacobi equation; their values at zero give the displayed constants.

For a variation of the initial point by a with terminal variation zero, the variational field is $C(s)a + J(s)b$, where $b = -J(T)^{-1}C(T)a$. The initial gradient of the divided action is the negative initial velocity. Differentiating it in a normal frame at x therefore gives $S(T) = J(T)^{-1}C(T)$. The last identity in (3.5) shows that this matrix is symmetric. Moreover, the first two identities give

$$J^*(C' - J'J^{-1}C) = J^*C' - J'^*C = -\text{Id}.$$

Consequently

$$\frac{d}{dT}(J^{-1}C) = J^{-1}(C' - J'J^{-1}C) = -J^{-1}J'^*,$$

including the order of the two inverse factors in (3.2).

For completeness, symplecticity also gives

$$\Phi(T)^{-1} = \begin{pmatrix} J'(T)^* & -J(T)^* \\ -C'(T)^* & C(T)^* \end{pmatrix}.$$

A Jacobi field for the reversed geodesic with initial position zero and initial derivative a has, in the forward parametrization, terminal data $(0, -a)$. Multiplication by this inverse matrix shows that its value at the other endpoint is $J(T)^*a$. The reversed Jacobi matrix at time T is thus $J(T)^*$. Dividing both matrices by T as in (3.1) proves (3.3); this argument does not require $J(T)$ to be invertible.

In the radial direction the Jacobi equation has the explicit solutions $J(s)v = sv$ and $C(s)v = v$. Thus $S(T)v = v/T$. For $v \in I(x)$, taking $T = 1$ gives (3.4); the case $v = 0$ is immediate. $\square$

3.2 Radial shortening at a conjugate endpoint

Lemma 3.2 (Uniform radial transversality). *There are constants $c_r > 0$ and $C_r < \infty$, depending only on (M, g) , such that for $x \in M$, $v \in G_x$, $0 < \delta \leq 1/2$, and $1 \leq j \leq n$,*

$$c_r(s_j(x, v) + \delta) \leq s_j(x, (1 - \delta)v) \leq C_r(s_j(x, v) + \delta). \quad (3.6)$$

There is also a positive definite form $Q_{x,v,\delta}$ on $T_x M$ such that, on putting

$$N_{x,\delta}(v) = -\delta(A_x((1 - \delta)v) - \text{Id}),$$

one has

$$\|N_{x,\delta}(v) - Q_{x,v,\delta}\| \leq C_r \delta, \quad (3.7)$$

$$c_r \frac{\delta}{s_j(x, v) + \delta} \leq \lambda_j^\downarrow(Q_{x,v,\delta}) \leq C_r \frac{\delta}{s_j(x, v) + \delta}. \quad (3.8)$$

In particular, uniformly on this set,

$$\|N_{x,\delta}(v)\| \leq C_r, \quad N_{x,\delta}(v) \succeq -C_r \delta \text{Id}, \quad N_{x,\delta}(v)v = 0. \quad (3.9)$$

No condition on the multiplicity of conjugacy is required.

Proof. The bundle

$$\mathcal{G} = \{(x, v) : x \in M, v \in G_x\}$$

is compact: its defining distance equality is closed, and $|v|$ is bounded by $\text{diam}(M)$. For each such vector consider $\gamma(s) = \exp_x(sv)$ on $[0, 1]$. Fix the splitting time $h = 1/4$ and allow the final time T to vary in $[1/2, 1]$. The prefix $h\gamma$ belongs to $I(x)$, since it extends by the factor $1/h = 4$ to the original minimizing ray. For the suffix, reverse the ray at $y_T = \gamma(T)$ and put

$$w_T = -(T - h)\dot{\gamma}(T), \quad a_T = \frac{T}{T - h} \geq \frac{4}{3}.$$

Then $\exp_{y_T} w_T = \gamma(h)$ and $d(y_T, \exp_{y_T}(a_T w_T)) = a_T |w_T|$, so $w_T \in I(y_T)$. The reversed suffix is therefore before cut; reversal preserves uniqueness and, by Lemma 3.1, nonconjugacy. The parameter family is compact and its image lies in the open injectivity domain. Thus both piece differentials and their inverses have uniform bounds. Their time lengths are bounded below by $1/4$.

Let C, J be the matrices from Lemma 3.1. Starting at time h , let $C_h(s), J_h(s)$ have the same initial data and solve the Jacobi equation along $\gamma(h + s)$. Smooth dependence on (x, v, T) and the preceding nonconjugacy show that

$$J(h), J(h)^{-1}, J_h(T - h), J_h(T - h)^{-1}, C(h), J'(h) \quad (3.10)$$

all have uniformly bounded norms. This compactness assertion also includes $v = 0$, for which $J(s) = s\text{Id}$.

Write $z = \gamma(h)$ and $y_T = \gamma(T)$. The Hessian in the joining point of the broken action

$$z' \mapsto \frac{c(x, z')}{h} + \frac{c(z', y_T)}{T - h}$$

at z is

$$D(T) = J'(h)J(h)^{-1} + J_h(T - h)^{-1}C_h(T - h). \quad (3.11)$$

The first term is the terminal Hessian of the first piece; the second is the source Hessian of the second piece. At $T = 1$, the broken action has a minimum at z . To see this directly, the triangle inequality and Cauchy–Schwarz give

$$\frac{d(x, z')^2}{2h} + \frac{d(z', y_1)^2}{2(1 - h)} \geq \frac{(d(x, z') + d(z', y_1))^2}{2} \geq \frac{d(x, y_1)^2}{2},$$

with equality at z . Hence $D(1) \succeq 0$.

Only the second term in (3.11) depends on T . Applying (3.2) with initial point z yields

$$D(1 - \delta) = D(1) + \delta G_\delta, \quad G_\delta = \frac{1}{\delta} \int_{1-\delta}^1 J_h(s - h)^{-1} J_h(s - h)^{-*} ds, \quad c\text{Id} \preceq G_\delta \preceq C\text{Id}. \quad (3.12)$$

The bounds on G_δ are precisely the bounds in (3.10): the smallest eigenvalue of the integrand is $\|J_h(s - h)\|^{-2}$, and its largest eigenvalue is $\|J_h(s - h)^{-1}\|^2$. Thus the crossing is positive on the whole space, including any null space of $D(1)$.

Jacobi propagation through the joining point gives the exact factorization

$$\begin{aligned} J(T) &= C_h(T-h)J(h) + J_h(T-h)J'(h) \\ &= J_h(T-h)D(T)J(h). \end{aligned} \quad (3.13)$$

This formula is also valid at $T = 1$ when either of the two full matrices is singular. The splitting point is kept fixed while T varies; differentiating a moving midpoint is unnecessary. Taking $h = 1/2$ gives the usual midpoint factorization on $T \in [3/4, 1]$. The choice $h = 1/4$ covers all the parameters in the statement with one set of bounds.

Write $\lambda_1(D) \leq \dots \leq \lambda_n(D)$ for the increasing eigenvalues of a positive semidefinite matrix D . The min-max characterization applied to (3.12) gives

$$\lambda_j(D(1)) + c\delta \leq \lambda_j(D(1-\delta)) \leq \lambda_j(D(1)) + C\delta.$$

Multiplication of a matrix on either side by a uniformly bounded invertible matrix changes each ordered singular value by at most a uniform multiplicative factor. Apply this fact to (3.13), use $(d\exp_x)_{Tv} = J(T)/T$, and note $1/2 \leq T \leq 1$. At $T = 1$, the singular values of $J(1)$ are comparable to the eigenvalues of $D(1)$; at $T = 1 - \delta$, they are comparable to the eigenvalues of $D(1 - \delta)$. These observations prove (3.6) with uniform constants.

We give the Hessian calculation as well, because it identifies the bounded term that must not be discarded at a radial eigenvector. Set

$$P = J(h)^{-1}C(h), \quad E = J(h)^{-*}.$$

The source-join mixed Hessian of the first divided action is $-J(h)^{-1} = -E^*$. For $T < 1$, eliminating the joining-point variation in the Hessian of the broken action gives

$$\frac{A_x(Tv)}{T} = P - E^*D(T)^{-1}E. \quad (3.14)$$

To verify the elimination, write the quadratic form for source variation a and joining variation b as $a^*Pa - 2a^*E^*b + b^*D(T)b$. Its unique stationary value in b is at $b = D(T)^{-1}Ea$ and has value $a^*(P - E^*D(T)^{-1}E)a$. This is the Hessian of the stationary composed action. Positivity of $D(T)$ follows from (3.12); the composed action is the smooth cost branch of the shortened minimizing ray.

For $T = 1 - \delta$, formula (3.14) gives

$$\begin{aligned} N_{x,\delta}(v) &= Q_{x,v,\delta} + \delta(\text{Id} - TP), \\ Q_{x,v,\delta} &= T\delta E^*D(T)^{-1}E. \end{aligned} \quad (3.15)$$

The second form is positive definite. The bounds on P, E, E^{-1} , the min-max inequalities just proved, and inversion of $D(T)$ give (3.7) and (3.8). They imply the norm and lower bounds in (3.9). The remaining identity follows from (3.4). In particular, for $v \neq 0$ the radial eigenvalue of $N_{x,\delta}(v)$ is exactly zero, while the radial singular value of $(d\exp_x)_v$ is one. These two facts are consistent because the comparison with $Q_{x,v,\delta}$ includes the bounded background $\delta(\text{Id} - TP)$. $\square$

Corollary 3.3 (Radial comparison of exponential determinants). *With a constant depending only on (M, g) , if $v \in G_x$ and $1/2 \leq \theta \leq 1$, then*

$$\sigma_x(\theta v) \geq c\sigma_x(v). \quad (3.16)$$

Suppose $y = \exp_x p$, $p \in G_x$, and $\bar{p} = -P_{x \rightarrow y}p$. If $l \in [1/2, 1)$ and $\bar{p}/l \in G_y$, then

$$\sigma_y(\bar{p}/l) \leq C\sigma_x(p). \quad (3.17)$$

If $l_+ \in [1, 2]$, then

$$\sigma_y(\bar{p}/l_+) \geq c\sigma_x(p). \quad (3.18)$$

Comparison (3.17) permits the longer reversed vector $\bar{p}/l$ to have a conjugate endpoint.

Proof. For $\theta < 1$, multiply the lower bounds in (3.6) with $\delta = 1 - \theta$ and discard the nonnegative terms δ . The case $\theta = 1$ is immediate. For (3.17), apply (3.16) to the longer vector $\bar{p}/l$ with shortening factor l , and then use (3.3). For (3.18), apply it to $\bar{p}$ with shortening factor $1/l_+$ and use the same reversal identity. Thus the direction of the two inequalities reflects, respectively, a longer and a shorter true reversed ray. $\square$

3.3 A uniform gain from an extra minimizing extension

Lemma 3.4 (Isotropic gain after pullback). *There is $c_p > 0$, depending only on (M, g) , with the following property. Let $y \in M$, $r \in T_y M$, and $1 < T_* \leq 2$ satisfy $T_* r \in G_y$. Put*

$$\gamma(s) = \exp_y(sr), \quad x = \gamma(1), \quad p = -P_{y \rightarrow x}r, \quad L = (d\exp_x)_p, \quad \tau = 1 + \frac{T_* - 1}{2}.$$

Then all costs appearing in

$$L^* \left[A_y(r) - \frac{1}{\tau} A_y(\tau r) \right] L \succeq c_p (T_* - 1) \text{Id} \quad (3.19)$$

are smooth. If the terminal ray at time T_* is nonconjugate, the same inequality holds with τ replaced by T_* and its chosen smooth branch. In particular, for $T_* = 1/l$ this latter inequality reads

$$L^* [A_y(r) - l A_y(r/l)] L \succeq c_p (1 - l) \text{Id}, \quad 1/2 \leq l < 1. \quad (3.20)$$

Without any uniformity assertion, the source Hessian of divided action decreases strictly in the positive definite order between any two nonconjugate times on a minimizing ray.

Proof. Every time less than T_* lies strictly before the cut time of this ray, so r and τr are in $I(y)$, and the reverse vector p is in $I(x)$. Let $J(s)$ be the Jacobi matrix based at y with unit initial derivative. By reversal,

$$L = J(1)^*. \quad (3.21)$$

Apply Lemma 3.2 to the minimizing vector $T_* r$, with shortening factor $1/T_*$. Since $\delta = (T_* - 1)/T_* \geq (T_* - 1)/2$, its lower bound gives

$$s_1(J(1)) = s_1(y, r) \geq c(T_* - 1), \quad \|J(1)^{-1}\| \leq \frac{C}{T_* - 1}. \quad (3.22)$$

There is also a uniform bound on $\|J'(s)\|$ for $0 \leq s \leq T_*$. Indeed, $|T_* r| \leq \text{diam}(M)$, $T_* \leq 2$, and smooth Jacobi evolution on this compact set of initial data bounds its fundamental matrices. Therefore, for $1 \leq s \leq \tau$,

$$\begin{aligned} \|J(s)J(1)^{-1}\| &\leq 1 + \|J(s) - J(1)\| \|J(1)^{-1}\| \\ &\leq 1 + C \frac{s - 1}{T_* - 1} \leq C. \end{aligned} \quad (3.23)$$

Write $S(s) = A_y(sr)/s$. Lemma 3.1 and (3.21) give

$$L^* [-S'(s)] L = J(1)J(s)^{-1}J(s)^{-*}J(1)^*. \quad (3.24)$$

The matrix on the right is $B_s B_s^*$ with $B_s = J(1)J(s)^{-1}$. Its smallest eigenvalue equals $\|B_s^{-1}\|^{-2} = \|J(s)J(1)^{-1}\|^{-2}$ and is thus bounded below by a uniform positive constant by (3.23). This uses the displayed order of the factors and does not commute any Jacobi matrices. Integration over $[1, \tau]$, whose length is $(T_* - 1)/2$, proves (3.19).

If the terminal branch is nonconjugate, S is defined through time T_* and $-S'(s)$ is positive definite on the remaining interval. Adding its integral over $[\tau, T_*]$ preserves the lower bound. Since $T_* - 1 = (1 - l)/l \geq 1 - l$, this gives (3.20). For arbitrary two nonconjugate times on a minimizing ray, integration of $-S'(s) = J(s)^{-1}J(s)^{-*} \succ 0$ proves strict decrease. When the true terminal point is conjugate, only the half-extension formula is used; no undefined terminal Hessian has been differentiated. $\square$

3.4 From transverse MTW to comparison along an affine segment

Lemma 3.5 (Uniform comparison along an extended segment). *Assume weak MTW. Fix $a_0 > 0$ and $B < \infty$. There is $c_s > 0$, depending only on $(M, g), a_0, B$, such that the following holds. Let $x \in M$, $p, q \in T_x M$, $q \neq 0$, satisfy*

$$p + tq \in G_x \quad (-a_0 \leq t \leq 1), \quad p \cdot q > 0, \quad \frac{|q|^2}{p \cdot q} \leq B. \quad (3.25)$$

Then, for every $1 \leq j \leq n$,

$$s_j(x, p) \geq c_s s_j(x, p + q). \quad (3.26)$$

In particular,

$$\sigma_x(p) \geq c_s^n \sigma_x(p + q). \quad (3.27)$$

Proof. If there is a configuration in the statement then $B > 0$. Set

$$a = \min\{a_0, (2B)^{-1}\}.$$

We use only the segment $[-a, 1]$. First suppose that all its points lie in $I(x)$. Write

$$p(t) = p + tq, \quad \beta = \frac{|q|^2}{p \cdot q}, \quad L(t) = \text{Id} + t \frac{qq^*}{p \cdot q}.$$

The eigenvalues of $L(t)$ are 1 on $q^\perp$ and $1 + t\beta$ on $\mathbb{R}q$. Hence

$$\|L(t)\| \leq 1 + B, \quad \|L(t)^{-1}\| \leq 2 \quad (-a \leq t \leq 1). \quad (3.28)$$

For $0 < \delta \leq 1/2$, put

$$N_\delta(t) = -\delta(A_x((1 - \delta)p(t)) - \text{Id}), \quad F_\delta(t) = L(t)^* N_\delta(t) L(t). \quad (3.29)$$

These matrices are uniformly bounded by Lemma 3.2 and (3.28), and $F_\delta(t) \succeq -C\delta \text{Id}$.

We next prove full matrix convexity of F_δ . For any fixed $z \in T_x M$, the vector

$$\xi = z - \frac{z \cdot q}{p \cdot q} p$$

is perpendicular to q . Moreover,

$$\xi - L(t)z = -\frac{z \cdot q}{p \cdot q} p(t).$$

The last vector belongs to the kernel of $N_\delta(t)$ by the radial identity. Symmetry consequently gives the exact equality

$$z^* F_\delta(t) z = \xi^* N_\delta(t) \xi. \quad (3.30)$$

Weak MTW means that $t \mapsto A_x((1 - \delta)p(t))(\xi, \xi)$ is concave whenever $\xi \perp q$. The negative sign in the definition of N_δ therefore makes the right side of (3.30) convex. Since this holds for every z , F_δ is convex in the Loewner order.

This argument remains uniform when $p \cdot q$ is small. The fixed test vector ξ itself might be large, but its value modulo the radial kernel is $L(t)z$. Equations (3.28) and (3.30) bound the scalar convex function by $C|z|^2$, independently of the size of ξ . For any compact interval $K \subset (-a, 1)$, a real convex function bounded in absolute value by $C|z|^2$ has slopes on K bounded by $2C|z|^2 / \text{dist}(K, \{-a, 1\})$: sandwich each secant slope between secants to the two endpoints. Applying this estimate to basis vectors and their sums and differences shows by polarization that every matrix entry of F_δ has a uniform Lipschitz bound on K .

To turn this bounded matrix convexity into a spectral comparison, we argue by contradiction. The spectral profile $\delta/(s_i + \delta)$ from (3.8) detects singular values below the radial scale: it stays positive when $s_i \ll \delta$ and tends to zero when $s_i \gg \delta$. At a scale separated from the singular values, the rank of a limiting matrix therefore counts those below the scale. A failing ordered singular-value ratio will provide a radial scale between the center and endpoint values. We choose it to separate asymptotically from every singular value at each fixed point of a countable dense set of segment parameters. The resulting positive semidefinite matrix limit has a uniform gap between zero and its positive eigenvalues, forcing constant interior rank. Convexity then prevents rank loss at the endpoint, whereas the failing ratio will force precisely such a loss.

Suppose that no uniform constant in (3.26) exists. There are configurations (x_k, p_k, q_k) satisfying the same a_0, B , an index j fixed after passing to a subsequence, and

$$\frac{s_j(x_k, p_k)}{s_j(x_k, p_k + q_k)} \longrightarrow 0. \quad (3.31)$$

For now all the segments are in their open injectivity domains, so the singular values are positive. Choose an orthonormal identification of each $T_{x_k}M$ with $\mathbb{R}^n$. No smooth choice in k is needed. Abbreviate

$$s_{i,k}(t) = s_i(x_k, p_k + tq_k), \quad R_k = \log \frac{s_{j,k}(1)}{s_{j,k}(0)} \longrightarrow \infty.$$

Let $\{t_1, t_2, \dots\}$ be a fixed countable dense subset of $(-a, 1)$, enlarged to include 0 and 1. We now choose one radial scale that separates every fixed member of this set. Take $\log \delta_k$ in the middle half of the interval

$$[\log s_{j,k}(0), \log s_{j,k}(1)].$$

Exclude the intervals of radius $R_k^{1/3}$ centered at

$$\log s_{i,k}(t_m), \quad 1 \leq i \leq n, \quad 1 \leq m \leq \lfloor R_k^{1/3} \rfloor.$$

Their total length is at most $2nR_k^{2/3}$, while the available middle interval has length $R_k/2$. For all sufficiently large k there is a remaining point, which we choose as $\log \delta_k$. This construction ensures

$$s_{j,k}(0)/\delta_k \longrightarrow 0, \quad s_{j,k}(1)/\delta_k \longrightarrow \infty, \quad (3.32)$$

$$\left| \log \frac{s_{i,k}(t_m)}{\delta_k} \right| \longrightarrow \infty \quad \text{for every fixed } i, m. \quad (3.33)$$

All singular values on $\mathcal{G}$ are uniformly bounded above. Since $\delta_k \leq s_{j,k}(1)e^{-R_k/4}$, we also have $\delta_k \rightarrow 0$. In particular these scales are eventually admissible in Lemma 3.2.

Form the matrices $F_k(t)$ in (3.29) using the k th segment and scale δ_k . Their uniform bounds and interior Lipschitz bounds give, by successive extraction on compact subintervals, a locally uniform limit $F(t)$ on $(-a, 1)$. Extract also a limit $E = \lim_k F_k(1)$. Both $F(t)$ and E are positive semidefinite, because their approximants are bounded below by $-C\delta_k \text{Id}$. The limit F is matrix convex and bounded.

To identify its possible ranks, use (3.7) to write

$$F_k(t) = L_k(t)^* Q_{x_k, p_k + tq_k, \delta_k} L_k(t) + \mathcal{E}_k(t), \quad \|\mathcal{E}_k(t)\| \leq C\delta_k.$$

The positive form in this expression has its i th decreasing eigenvalue between uniform multiples of

$$\frac{\delta_k}{s_{i,k}(t) + \delta_k}, \quad (3.34)$$

by (3.8) and (3.28). At a fixed dense point t_m , put $\eta_k = \exp(-R_k^{1/3})$. For all sufficiently large k , depending on m , (3.33) says that $s_{i,k}(t_m)/\delta_k$ is at most η_k or at least η_k^{-1} . Weyl's eigenvalue inequality therefore gives one of the two alternatives

$$\lambda_i^\downarrow(F_k(t_m)) \geq \frac{c_0}{1 + \eta_k} - C\delta_k, \quad \text{or} \quad -C\delta_k \leq \lambda_i^\downarrow(F_k(t_m)) \leq C_0\eta_k + C\delta_k,$$

with $c_0, C_0 > 0$ uniform in i, m, k . Since $F_k(t_m)$ already converges, its ordered eigenvalues converge. Their limits are therefore either zero or at least c_0 , even if the alternatives initially alternate. No simultaneous rate at all dense points is required. Set $c_* = c_0/2$.

The ordered eigenvalues of a symmetric matrix are continuous in its entries. Density of the set $\{t_m\}$ and continuity of $F(t)$ therefore give

$$\text{spec } F(t) \subset \{0\} \cup [c_*, C_*] \quad (-a < t < 1) \quad (3.35)$$

after decreasing c_* once if necessary. Indeed, an eigenvalue strictly between zero and c_* at an interior point would remain in that interval in a neighborhood and contradict the dense-set conclusion. The rank is now locally constant, because a continuous eigenvalue cannot pass from zero to a positive value without entering that gap. It is consequently constant on the connected interval $(-a, 1)$.

The endpoint also has to be treated; constancy of the interior rank alone would not suffice. For every fixed vector z , the bounded scalar convex function $z^*F(t)z$ has a finite limit as $t \uparrow 1$. Polarization yields a matrix limit $F(1-)$. The gap (3.35) persists in this limit, so its rank equals the interior rank. For fixed $s < t < 1$, matrix convexity of F_k says

$$F_k(t) \preceq \frac{1-t}{1-s} F_k(s) + \frac{t-s}{1-s} F_k(1).$$

First let $k \rightarrow \infty$, then let $t \uparrow 1$; this gives

$$F(1-) \preceq E. \quad (3.36)$$

Positive semidefinite order implies $\text{rank } E \geq \text{rank } F(1-)$, since the kernel of E is contained in the kernel of $F(1-)$.

On the other hand, the first inequality in (3.32) and the ordering of singular values make the first j eigenvalues in (3.34) at $t = 0$ bounded away from zero. Thus $\text{rank } F(0) \geq j$. The second inequality makes all eigenvalues with decreasing index $i \geq j$ tend to zero at $t = 1$, so $\text{rank } E \leq j - 1$.

This contradicts the interior rank constancy and (3.36). The uniform ordered comparison is proved for open segments.

Finally, replace a closed segment in (3.25) by $(1 - \varepsilon)(p + tq)$, with $0 < \varepsilon < 1$. It lies in $I(x)$ by radial shortening. Its extension interval and the ratio $|q|^2/(p \cdot q)$ are unchanged, so the same constant applies. Letting $\varepsilon \downarrow 0$ and using continuity of the singular values proves (3.26) on G_x . Multiplication over j proves (3.27). Every bound in the contradiction argument depended only on $(M, g), a_0, B$, which establishes the stated uniformity. $\square$

3.5 Jacobian comparison for a barycentric configuration

Corollary 3.6 (Barycentric configurations). *Assume weak MTW and the convexity of G_x established in Corollary 2.6. Fix $\mu \in (0, 1)$ and positive constants κ, L_0 . Suppose*

$$p = \sum_{i=1}^N m_i p_i, \quad p_i \in G_x, \quad m_i > 0, \quad \sum_i m_i = 1,$$

and for one index i there are $D > 0$, a unit vector e , and $t_i > 0$ such that

$$p_i = p + t_i e, \quad h := p \cdot e > 0, \quad h^2 \geq \kappa D, \quad |p_i|^2 - |p|^2 \leq L_0 D, \quad m_i \geq \mu. \quad (3.37)$$

Then

$$\sigma_x(p) \geq c \sigma_x(p_i), \quad (3.38)$$

where $c > 0$ depends only on $(M, g), \mu, \kappa, L_0$.

Proof. Set $q = t_i e$. Expansion of the squared norms in (3.37) gives

$$2ht_i + t_i^2 \leq L_0 D, \quad \frac{|q|^2}{p \cdot q} = \frac{t_i}{h} \leq \frac{L_0 D}{2h^2} \leq \frac{L_0}{2\kappa}.$$

Because $q \neq 0$, $m_i < 1$. The remaining barycenter belongs to G_x by its convexity and is

$$\frac{1}{1 - m_i} \sum_{j \neq i} m_j p_j = p - \frac{m_i}{1 - m_i} q.$$

The coefficient $m_i/(1 - m_i)$ is at least $m_i \geq \mu$. Convexity therefore puts $p + sq$ in G_x for every $s \in [-\mu, 1]$. Lemma 3.5 applies with $a_0 = \mu$ and $B = L_0/(2\kappa)$ and proves the result. No upper bound on $m_i/(1 - m_i)$ is needed: a shorter common opposite extension is enough. $\square$

4 Sections, scales, and a quantitative separation

Throughout this section the manifold and the density bounds are fixed. We use the negative-cost convention

$$f^c(x) = \max_{y \in M} \{-c(x, y) - f(y)\}.$$

Thus a cost-dual pair satisfies $u = v^c$, $v = u^c$, and $u(x) + c(x, y) + v(y) \geq 0$. Let $\mathcal{C}$ denote the class of these pairs for all admissible density pairs, with any fixed additive normalization. All members of $\mathcal{C}$ have a common Lipschitz bound, and their oscillations are uniformly bounded. None of the arguments below uses derivatives of the densities.

4.1 A scalar quantity controlling both transport directions

For $(u, v) \in \mathcal{C}$ and $r \geq 0$, put

$$S_r(x; u, v) = \{y \in M : u(x) + c(x, y) + v(y) \leq r\}, \quad F(r) = \sup_{(u, v) \in \mathcal{C}, x \in M} \text{osc}_{S_r(x; u, v)} v.$$

The sets S_r are nonempty and compact. The function F is bounded and nondecreasing.

Lemma 4.1 (The power reduction). *If there are $\beta > 0$, $C_0 < \infty$, and $r_0 > 0$ such that $F(r) \leq C_0 r^\beta$ for $0 < r < r_0$, then all the optimal maps in the fixed density class have homeomorphic representatives with a common Hölder estimate in both directions. One may take any exponent no larger than $\frac{1}{2} \min\{\beta, 1\}$.*

Proof. Write

$$h_x(p) = u(x) + \frac{1}{2}|p|^2 + v(\exp_x p), \quad p \in G_x.$$

By Corollary 2.7, the lifted section $\{p \in G_x : h_x(p) \leq r\}$ is convex. If p, q belong to it and $m = (p + q)/2$, the squared-norm identity gives

$$\begin{aligned} \frac{1}{8}|p - q|^2 &= v(\exp_x m) - \frac{v(\exp_x p) + v(\exp_x q)}{2} + \frac{h_x(p) + h_x(q)}{2} - h_x(m) \\ &\leq F(r) + r. \end{aligned} \tag{4.1}$$

The exponential map has a common Lipschitz bound on the compact bundle of minimizing velocities. Hence the target diameter of a section is at most $C\sqrt{F(r) + r}$.

Let (x, y) and (x', y') be contacts of the same dual pair. The common Lipschitz bounds for u and c imply

$$0 \leq u(x) + c(x, y') + v(y') \leq Cd(x, x').$$

Thus y, y' belong to a section at x of that height. For sufficiently small $d(x, x')$, (4.1) gives

$$d(y, y') \leq Cd(x, x')^{\frac{1}{2} \min\{\beta, 1\}}.$$

In particular, letting the section height tend to zero shows that every pole has exactly one contact endpoint. The bound proves that this endpoint depends continuously on the pole. The class also contains the reversed pair (v, u) after renormalization, which does not affect its section oscillations. The same reasoning therefore gives a unique continuous contact in the reverse direction. The dual identities give a contact above every point of either copy of M , and the two resulting maps are inverse. They agree almost everywhere with the optimal maps by optimality and uniqueness. Finally, compactness absorbs the remaining, larger distances into the same uniform constant. $\square$

We shall contradict failure of a power bound for F . The following elementary selection makes the required quantifiers explicit.

Lemma 4.2 (Logarithmic plateaus). *Suppose that F has no power upper bound near zero. There are $r_j < R_j \rightarrow 0$, pairs $(u_j, v_j) \in \mathcal{C}$, poles x_j , and numbers a_j, D_j with $D_j > 0$ such that*

$$\min_{S_{r_j}(x_j; u_j, v_j)} v_j = a_j, \quad \max_{S_{r_j}(x_j; u_j, v_j)} v_j = a_j + D_j, \tag{4.2}$$

$$\frac{r_j}{R_j} \rightarrow 0, \quad F(R_j) \leq (1 + \delta_j)D_j, \quad \delta_j \rightarrow 0, \quad \frac{|\log D_j|}{|\log R_j|} \rightarrow 0. \tag{4.3}$$

Consequently $R_j/D_j^k \rightarrow 0$ for every fixed $k > 0$.

Proof. If F vanished at a positive scale, monotonicity would already give a power bound below that scale. Thus $F(r) > 0$. Fix $C_F \geq \max\{1, \sup_r F(r)\}$ and set

$$f(s) = \log \frac{C_F}{F(e^{-s})}.$$

This function is nonnegative and nondecreasing. Failure of every power bound implies $\liminf_{s \rightarrow \infty} f(s)/s = 0$; otherwise $f(s) \geq \beta s$ eventually for some $\beta > 0$. Choose $N_j \rightarrow \infty$ with $f(N_j) = o(N_j)$, and divide $[N_j/2, N_j]$ into

$$m_j = \left\lceil \sqrt{N_j \max\{1, f(N_j)\}} \right\rceil$$

equal intervals. Their length $N_j/(2m_j)$ tends to infinity. Since their total f -increment is at most $f(N_j)$, some interval $[A_j, B_j]$ has increment at most $f(N_j)/m_j = o(1)$. Put $R_j = e^{-A_j}$ and $r_j = e^{-B_j}$. Then

$$\log(R_j/r_j) = B_j - A_j \rightarrow \infty, \quad \frac{F(R_j)}{F(r_j)} = e^{f(B_j) - f(A_j)} \rightarrow 1.$$

Choose an actual section with oscillation $D_j \geq (1 - 1/j)F(r_j)$, discarding the first few indices. Its extrema are attained, which gives (4.2). Moreover $A_j \geq N_j/2$ and $f(B_j) \leq f(N_j) = o(N_j)$, so $|\log D_j| = o(A_j)$. This proves (4.3). Finally

$$\log(R_j/D_j^k) = -A_j - k \log D_j = -A_j + o(A_j) \rightarrow -\infty.$$

□

We suppress the index j from now on. All asymptotic statements refer to the sequence in Lemma 4.2, after the templates below have been fixed. It is not necessary that D tend to zero. Set

$$b_- = r(R/r)^{1/3}, \quad b_+ = r(R/r)^{2/3}. \quad (4.4)$$

Then $r \ll b_- \ll b_+ \ll R$. Uniformly for $b \in [b_-, b_+]$ and $0 \leq \tau \leq 8b/D$,

$$\frac{r}{b} \rightarrow 0, \quad \frac{b}{R} \rightarrow 0, \quad \frac{b}{D^k} \rightarrow 0 \text{ } (k > 0 \text{ fixed}), \quad \frac{\tau}{D} \rightarrow 0, \quad \frac{\tau^2}{b} \rightarrow 0. \quad (4.5)$$

For example, $\tau^2/b \leq 64b/D^2$. These estimates, rather than an additional restriction on R/r , will control the Taylor errors.

4.2 Templates and the order of choices

At the normalized level $s = (v - a)/D$, we will use the two modifications $v + bB_c(s)$ and $v + bB_o(s)$. The first is the center datum; we compare it with the double cost transform of the second. A positive excess in this comparison will produce an interior maximum. We first fix the center profile and a cap on the outer values, then choose the outer curvature parameter K and its profile, and only afterward take the gap scales small. The construction below makes that order explicit.

We use the fixed numerical choices

$$\varepsilon_0 = 2^{-10}, \quad c_0 = 2^{-10}, \quad c_1 = c_0/4. \quad (4.6)$$

The constants are deliberately smaller than needed.

The value constraints below serve the envelope comparison: the center profile stays near one on the lower half of the selected oscillation, whereas the outer profile vanishes at level one and places a barrier on intermediate levels. We also prescribe strong outer concavity where the outer value is bounded by a fixed cap. That derivative condition will enter the endpoint lower tests in Section 5; the height bound below keeps the constants in the switch and excess bounds independent of K .

Lemma 4.3 (Smooth templates). *There is a fixed smooth function $B_c : \mathbb{R} \rightarrow (-1, 1)$ with bounded derivatives such that*

$$B'_c < 0 \text{ on } \mathbb{R}, \quad B_c(s) \geq 1 - \varepsilon_0 \quad (s \leq 1/2), \quad B_c(s) < 0 \quad (s \geq 3/4).$$

Given a cap $M_0 \geq 2$ and $K \geq 1$, there are $0 < \eta \leq 1/128$ and a smooth bounded function $B_o \geq 0$, with bounded derivatives, satisfying

$$B_o(s) \geq 1 \quad (s \leq 1 - \eta), \quad B_o(1) = 0, \quad 1 \leq B_o(0) \leq 1 + \varepsilon_0, \quad (4.7)$$

$$B_o(s) \geq B_o(0) + 1 \quad (1/16 \leq s \leq 1 - \eta), \quad (4.8)$$

$$B'_o(s) > 0, \quad B''_o(s) \leq -K \quad \text{if } -2\eta \leq s \leq 1 - \eta \text{ and } B_o(s) \leq M_0. \quad (4.9)$$

One can arrange $\|B_o\|_\infty \leq 2M_0 + 8$, independently of K , and make η arbitrarily smaller if desired.

Proof. Choose γ so large that $2/(1 + e^{\gamma/8}) \leq \varepsilon_0$, and set

$$B_c(s) = 1 - \frac{2}{1 + e^{-\gamma(s-5/8)}}.$$

This has all the asserted properties. In particular, $|B'_c|$ has a positive minimum on every fixed compact interval.

For the outer function let

$$\beta(z) = \begin{cases} e^{-1/z}, & z > 0, \\ 0, & z \leq 0, \end{cases} \quad \chi(z) = \frac{\beta(z)}{\beta(z) + \beta(1-z)}.$$

Thus χ is a smooth step, equal to zero on $(-\infty, 0]$ and to one on $[1, \infty)$. Put

$$\kappa = K + 1, \quad A = M_0 + 3, \quad L = 128A\kappa, \quad h_0 = \frac{1}{128\kappa}, \quad 0 < \eta \leq \min \left\{ \frac{1}{128}, \frac{\varepsilon_0}{24L} \right\}.$$

Define

$$\begin{aligned} w(s) &= (L - \kappa s) \chi\left(\frac{s + 3\eta}{\eta}\right) \left[1 - \chi\left(\frac{s - h_0}{h_0}\right)\right], \\ J(s) &= 1 + \frac{\varepsilon_0}{2} + \int_0^s w(q) dq, \\ B_o(s) &= J(s) \left[1 - \chi\left(\frac{s - (1 - \eta)}{\eta}\right)\right]. \end{aligned}$$

The function w is nonnegative and supported in $[-3\eta, 2h_0]$. Also

$$\int_{-3\eta}^0 w \leq 3\eta(L + 3\kappa\eta) \leq 6\eta L \leq \varepsilon_0/4, \quad \int_0^{2h_0} w \leq 2Lh_0 = 2A.$$

It follows that

$$1 + \varepsilon_0/4 \leq J(s) \leq 1 + \varepsilon_0/2 + 2A.$$

On $[-2\eta, h_0]$, one has $J' = L - \kappa s > 0$ and $J'' = -\kappa$. Moreover

$$J(h_0) - J(0) = A - \frac{1}{32768\kappa} > M_0 + 1.$$

Since J is nondecreasing, the condition $B_o(s) \leq M_0$ on $[-2\eta, 1 - \eta]$ forces $s < h_0$; here $B_o = J$, proving (4.9). The function J is constant beyond $2h_0 \leq 1/64$, so (4.8) also follows. The final factor is identically one up to $1 - \eta$, and decreases to zero only in $(1 - \eta, 1)$. This proves the remaining assertions, including smoothness at all joins and the uniform height bound. $\square$

The cap M_0 will be fixed independently of K by Lemma 4.9; the choice $M_0 = 24(n+1)/c_1$ suffices. The center template is already fixed. After fixing M_0 , choose K and the outer template, and only then take the scales small. Besides (4.5), this order gives

$$\delta_j = o(\eta), \quad \frac{b}{D} \max\{\|B'_o\|_\infty, \|B'_c\|_\infty\} \rightarrow 0, \quad \frac{b}{D^2} \max\{\|B''_o\|_\infty, \|B''_c\|_\infty\} \rightarrow 0. \quad (4.10)$$

In particular, both scalar modifications used below are increasing. Constants in intermediate local estimates may depend on a fixed template, but the constants c_0, c_1 and the cap do not depend on K .

4.3 The first switch

Fix a selected section with extreme endpoints y_0, y_1 , so that $v(y_0) = a$ and $v(y_1) = a + D$. Choose minimizing logs $p_0, p_1 \in G_{x_0}$ to these endpoints, and put $P = |p_0|^2$. For every target write

$$s_y = \frac{v(y) - a}{D}, \quad w_{o,b}(y) = v(y) + bB_o(s_y), \quad w_{c,b}(y) = v(y) + bB_c(s_y), \quad g_b = w_{o,b}^c.$$

Let $H_o = \|B_o\|_\infty$. Since $B_o \geq 0$,

$$u - bH_o \leq g_b \leq u. \quad (4.11)$$

We will move the pole toward y_0 . Initially the outer modification lowers the level-zero mountain by about b , while leaving y_1 unpenalized. As long as mountains with $s_y \geq 1 - \eta$ dominate, their minimizing logs are shorter than the log to y_0 , so they grow more slowly along this motion. The barrier excludes intermediate-level winners, forcing a switch with both a low and a high active endpoint. Their two logs will then provide a supported intermediate log with a prescribed squared norm; the next subsection converts this construction into a positive excess.

We record the uniform distance estimate used in the proof. Splitting a minimizing geodesic at its midpoint gives a smooth upper support for the squared-distance cost with a uniformly bounded Hessian. Consequently, for any minimizing log q from x to y ,

$$c(\exp_x z, y) \leq c(x, y) - \langle q, z \rangle + C_g |z|^2 \quad \text{for } |z| \text{ sufficiently small,} \quad (4.12)$$

with one geometric constant and radius. Along a smooth curve, the one-sided derivative of $-c(\cdot, y)$ is the maximum of the pairings with its minimizing logs. In particular, whenever this restriction is differentiable,

$$\frac{d}{d\tau} [-c(x_\tau, y)] \leq d(x_\tau, y) |\dot{x}_\tau|. \quad (4.13)$$

These assertions hold also at a cut endpoint. The same upper derivative bound applies to an active mountain of a maximum: at a time when the maximum is differentiable, a semiconvex active mountain has the same derivative. Indeed its one-sided derivatives straddle that derivative in the opposite order to their semiconvex ordering, and hence coincide. The restrictions in question are Lipschitz, so these differential inequalities can be integrated.

Lemma 4.4 (First switch with absolute level control). *For all sufficiently small selected scales, uniformly for $b \in [b_-, b_+]$, there is*

$$\frac{b}{32D} \leq \tau_b \leq \frac{8b}{D} \quad (4.14)$$

such that, at $x_b = \exp_{x_0}(\tau_b p_0)$, the function g_b has active endpoints y_- and y_+ with

$$-2\eta \leq s_{y_-} < 1/16, \quad s_{y_+} \geq 1 - \eta.$$

The original endpoints y_0, y_1 and every active endpoint at this pole lie in one original section $S_{Cb}(x_b; u, v)$, where C depends only on the geometry and H_o . In particular every active level is in $[-\delta_j, 1 + \delta_j]$. For every minimizing $\log q_i$ to an active endpoint,

$$|q_i|^2 = P - 2Ds_{y_i} + O_{H_o}(\tau_b + b) = P - 2Ds_{y_i} + o(D). \quad (4.15)$$

The constants in (4.14) are independent of K and of the height and derivatives of the outer template.

Proof. Set $x_\tau = \exp_{x_0}(\tau p_0)$ for $0 \leq \tau \leq T := 8b/D$, and let Π_τ be parallel transport along this geodesic. For small scales $T < 1/2$. The left cost is exact:

$$f_y(\tau) := -c(x_\tau, y) - v(y), \quad f_0(\tau) = -\frac{1}{2}(1 - \tau)^2 P - a, \quad q_0(\tau) = (1 - \tau)\Pi_\tau p_0.$$

The initial gaps at y_0, y_1 lie in $[0, r]$, whence

$$P - |p_1|^2 = 2D + O(r), \quad |p_0 - p_1|^2 \leq 8(D + r), \quad P - \langle p_0, p_1 \rangle \leq 6D \quad (4.16)$$

for small scales; the second inequality is (4.1) applied using the actual oscillation D of this section. In particular $P \geq D$.

First exclude the middle levels. If $1/16 \leq s_y \leq 1 - \eta$, put

$$h_y(\tau) = f_y(\tau) - bB_o(s_y) - f_0(\tau) + bB_o(0).$$

Initially $h_y(0) \leq r - b$. Wherever $h_y \geq -b/2$, the identity relating values and squared distances gives, for every minimizing $\log q_y$ at the current pole,

$$|q_y|^2 - |q_0|^2 = -2Ds_y - 2b(B_o(s_y) - B_o(0)) - 2h_y \leq -D/8 - b.$$

Equations (4.13) and $\dot{x}_\tau = \Pi_\tau p_0$ imply $h'_y \leq -D/16$ at almost every such time. The positive part of $h_y + b/2$ is therefore nonincreasing and starts at zero. Thus $h_y \leq -b/2$ throughout $[0, T]$. No middle-level mountain can be active.

Let G_L and G_R be the maxima of $f_y - bB_o(s_y)$ over the compact sets $s_y \leq 1/16$ and $s_y \geq 1 - \eta$, respectively. Both are continuous, and the preceding paragraph shows $g_b(x_\tau) = \max\{G_L(\tau), G_R(\tau)\}$. Initially $G_L(0) \leq u(x_0) - b$ and $G_R(0) \geq f_1(0) \geq u(x_0) - r$. While $G_R > G_L$, every active endpoint is right and comparison with the left modified mountain gives

$$|q_y|^2 \leq |q_0|^2 - 2D(1 - \eta) + 2bB_o(0) \leq |q_0|^2 - D.$$

By rationalizing the difference of the square roots in (4.13),

$$\frac{d}{d\tau} \{g_b(x_\tau) - f_0(\tau)\} \leq -D/2 \quad \text{almost everywhere while } G_R > G_L.$$

This difference starts at most r and is always at least $-bB_o(0) \geq -2b$. It cannot obey the displayed inequality throughout $[0, T]$. There is therefore a first equality time $\tau_b \leq T$. At that time both maxima are attained, and the middle-level exclusion gives $s_{y_-} < 1/16$. Throughout the preceding interval, including its endpoint,

$$-bB_o(0) \leq g_b(x_\tau) - f_0(\tau) \leq r. \quad (4.17)$$

We next anchor the absolute levels. By (4.11) and (4.17), $u(x_\tau) - f_0(\tau) \leq bH_o + r$. The initial-cost upper support and (4.16) give

$$f_1(\tau) \geq f_0(0) - r + (P - 6D)\tau - C_g \tau^2 =: \mathcal{L}(\tau). \quad (4.18)$$

Since $f_0(\tau) = f_0(0) + P\tau - P\tau^2/2$, it follows that $f_0 - f_1 \leq r + 6D\tau + C\tau^2 = O(b)$ on this interval. Thus both y_0, y_1 lie in an original section of height Cb . For any active endpoint y ,

$$u(x_\tau) + c(x_\tau, y) + v(y) = u(x_\tau) - g_b(x_\tau) - bB_o(s_y) \leq bH_o.$$

All these endpoints belong to the same section of height $Cb < R$. Its oscillation is at most $(1 + \delta_j)D$ and it contains levels $a, a + D$. Hence every active level lies in $[-\delta_j, 1 + \delta_j]$, proving in particular the required lower bound for s_{y_-} .

To prove the lower time estimate, keep the winning left endpoint y_- fixed, and put

$$k(\tau) = f_{y_-}(\tau) - bB_o(s_{y_-}) - \mathcal{L}(\tau).$$

One has $k(0) \leq 2r - b < -b/2$, while $k(\tau_b) \geq 0$, since $g_b(x_{\tau_b}) \geq f_1(\tau_b)$. Let θ be the last time before τ_b at which $k = -b/2$. Then $k \geq -b/2$ on $[\theta, \tau_b]$. On this interval, using $s_{y_-} \geq -2\eta$ and $B_o \geq 0$, the inequality defining this last strip gives

$$\begin{aligned} |q_{y_-}|^2 - |q_0|^2 &\leq 4\eta D + 2r + 12D\tau + 2C_g\tau^2 + b \\ &\leq D \end{aligned}$$

for all sufficiently small scales. This uses only (4.5); in particular $\tau/D \rightarrow 0$. Rationalizing once more gives $f'_{y_-} \leq (1 - \tau)P + 2D$ almost everywhere in the strip. Since $\mathcal{L}' = P - 6D - 2C_g\tau$, we have $k' \leq 10D$ there for small scales. Therefore

$$b/2 \leq k(\tau_b) - k(\theta) \leq 10D(\tau_b - \theta),$$

which implies the stated weaker lower bound $\tau_b \geq b/(32D)$. No upper bound for the template or its derivatives occurs in this last estimate.

Finally, if y_i is active at the switch, the exact equality of its modified value with g_b gives

$$|q_i|^2 = (1 - \tau_b)^2 P - 2Ds_{y_i} - 2bB_o(s_{y_i}) - 2\{g_b(x_b) - f_0(\tau_b)\}.$$

Equation (4.17) and (4.5) prove (4.15). □

4.4 Positive excess and an interior maximum

Write $\hat{w}_b = g_b^c = w_{o,b}^{cc}$ and

$$E_b(y) = w_{c,b}(y) - \hat{w}_b(y), \quad E(b) = \max_{y \in M} E_b(y).$$

Order reversal of one cost transform and the double-transform inequality give

$$v \leq \hat{w}_b \leq w_{o,b}, \quad E_b(y) \leq bB_c(s_y) \leq b. \quad (4.19)$$

Lemma 4.5 (Uniform positive excess). *For all sufficiently small selected scales and all $b \in [b_-, b_+]$,*

$$c_0 b \leq E(b) \leq b, \quad (4.20)$$

with the constant in (4.6), independently of K .

Proof. Use the two active logs at the switch supplied by Lemma 4.4. Equation (4.15) and their level bounds show that their squared norms lie on opposite sides of $P - D/2$. By continuity along their segment, choose $p \in \partial g_b(x_b)$ with

$$|p|^2 = P - D/2.$$

Theorem 2.5 says that p is minimizing and that $y = \exp_{x_b} p$ is a contact endpoint for g_b . Thus

$$E_b(y) = v(y) + bB_c(s_y) + c(x_b, y) + g_b(x_b).$$

Minorizing g_b by the left modified mountain gives the direct bound

$$E_b(y) \geq D(s_y - 1/4) + P\tau_b - P\tau_b^2/2 + b\{B_c(s_y) - B_o(0)\}. \quad (4.21)$$

If $s_y > 1/2$, the right side is at least $D/4 - 3b$, which is larger than c_0b for small scales.

It remains to consider $s_y \leq 1/2$. The original support inequality at x_0 yields

$$v(y) - a \geq c(x_0, y_0) - c(x_0, y) - r.$$

Apply (4.12) at x_b with backward displacement $-\tau_b \Pi_{\tau_b} p_0$ to get

$$c(x_b, y) - c(x_0, y) \geq -\tau_b \langle p, \Pi_{\tau_b} p_0 \rangle - C\tau_b^2.$$

The corresponding cost difference for y_0 is exactly $-P\tau_b + P\tau_b^2/2$. Therefore

$$\frac{E_b(y)}{b} \geq B_c(s_y) - B_o(0) + (P - \langle p, \Pi_{\tau_b} p_0 \rangle) \frac{\tau_b}{b} - \frac{r}{b} - C \frac{\tau_b^2}{b}. \quad (4.22)$$

The norm choice gives

$$P - \langle p, \Pi_{\tau_b} p_0 \rangle = \frac{P - |p|^2 + |p - \Pi_{\tau_b} p_0|^2}{2} \geq D/4.$$

The middle term in (4.22) is consequently at least $1/128$. The template difference is at least $-2\varepsilon_0$, and the last two terms tend to zero uniformly by (4.5). The numerical choices (4.6) give the asserted lower bound. The upper bound is (4.19). $\square$

To obtain an interior maximum, we now subtract a penalty from the excess. A small linear term preserves its positive lower bound, and a quadratic term makes the upper amplitude boundary unfavorable. Define

$$\pi(b) = \frac{c_0}{4}b + \frac{2b^2}{b_+}. \quad (4.23)$$

In particular, $\pi'(b) \geq c_1$.

Lemma 4.6 (Regularized interior maxima). *Fix a sufficiently small selected scale. For every $t < 1$ sufficiently close to one, the continuous function*

$$\Phi_t(b, z) = Q_t g_b(z) + Q_{1-t} w_{c,b}(z) - \pi(b), \quad (b, z) \in [b_-, b_+] \times M,$$

has a positive global maximum with b in the interior of its interval. Both spatial terms are $C^{1,1}$, and both minimizing points are unique. If x and y are these minimizing points at a maximum, there is $p \in \partial g_b(x) \cap G_x$ such that

$$z = \exp_x(tp), \quad y = \exp_x p, \quad Q_t g_b(z) + Q_{1-t} w_{c,b}(z) = E_b(y) > 0. \quad (4.24)$$

Moreover (x, y) is before cut.

Proof. At $b = b_+$, (4.20) makes $E(b) - \pi(b)$ negative. At $b = b_-$, it is at most b_- . At $b_* = c_0 b_+/8$, which belongs to the interval for small scales,

$$E(b_*) - \pi(b_*) \geq c_0^2 b_+/16.$$

Since $b_-/b_+ \rightarrow 0$, these inequalities give a strictly positive interior maximum with a positive separation from both boundary values.

For a fixed scale and b in its compact interval, $Q_t g_b \rightarrow Q_1 g_b = -\hat{w}_b$ uniformly as $t \uparrow 1$. This follows directly from the uniform bound for $|c/t - c|$. The functions $w_{c,b}$ have a common Lipschitz constant L_* on that interval, and

$$w_{c,b} - \frac{1}{2} L_*^2 (1-t) \leq Q_{1-t} w_{c,b} \leq w_{c,b}.$$

Thus $\Phi_t \rightarrow E_b - \pi$ uniformly in (b, z) , preserving the positive interior maximum when t is sufficiently close to one.

Proposition 2.8 gives the asserted regularity and unique pole for $Q_t g_b$. To check the other term, the scalar function $q \mapsto q + bB_c((q-a)/D)$ has positive derivative, and its first two derivatives are bounded at the fixed scales. Composing a Lipschitz semiconvex function with this function preserves semiconvexity: the lower Taylor support for v , its Lipschitz bound, and the ordinary second-order Taylor inequality for the scalar function give a common quadratic lower support for $w_{c,b}$. For completeness, a minimizer in $Q_\epsilon w_{c,b}(z)$ has $d(z, y) \leq 2L_*\epsilon$, by comparison with $y = z$. For small ϵ , all minimizers therefore lie in one convex normal ball, where the Hessian of $c(z, \cdot)/\epsilon$ dominates the negative semiconvexity bound. Strict convexity gives uniqueness. In one fixed coordinate chart, let K_* be the semiconvexity constant and let μ/ϵ be a lower Hessian bound for the smooth cost term. The subgradient inequalities at the minimizers y, y' for nearby z, z' and the mixed cost derivative bound give

$$(\mu/\epsilon - K_*)|y - y'|^2 \leq (C/\epsilon)|z - z'| |y - y'|.$$

For small ϵ this proves local Lipschitz dependence on z . The envelope gradient formula then gives $C^{1,1}$ regularity. This is uniform in b on the fixed interval.

At a maximum, the two spatial gradients sum to zero. Their envelope formulas show that the geodesic from x to z , continued for the remaining time $1 - t$ with the same velocity, reaches y . The pole minimum gives $p \in \partial g_b(x)$. By Theorem 2.5, this full vector is minimizing and its endpoint is a support for g_b . Splitting its energy at time t proves (4.24) exactly.

Finally put $l = 1 + bB'_c(s_y)/D \in (0, 1)$. The short final minimization gives its backward velocity in $\partial w_{c,b}(y) = l\partial v(y)$; the equality follows from the increasing scalar chain rule, which follows in both directions by Taylor expansion and inversion of that scalar function. Dividing by l gives a minimizing velocity by Theorem 2.5. The backward velocity to x is a strict radial shortening of this vector, and is therefore before cut. $\square$

4.5 The parameter inequality for every representation

Lemma 4.7 (The parameter derivative). *At any maximum in Lemma 4.6, let*

$$p = \sum_{i=1}^N m_i p_i, \quad m_i > 0, \quad \sum_i m_i = 1,$$

be any finite representation by minimizing logs active for the displayed representation $g_b = w_{o,b}^c$ at its minimizing pole x . Thus $g_b(x) + c(x, \exp_x p_i) + w_{o,b}(\exp_x p_i) = 0$. Write $y_i = \exp_x p_i$ and $s_i = (v(y_i) - a)/D$, and let $s = (v(y) - a)/D$ be the center level. Then

$$B_c(s) - \sum_i m_i B_o(s_i) \geq \pi'(b) \geq c_1. \quad (4.25)$$

Proof. Fix the prefix time and the spatial maximum point z , and abbreviate $A(b) = Q_t g_b(z)$ and $C(b) = Q_{1-t} w_{c,b}(z)$. Decrease b by $h > 0$, and let x_h be the new pole for $A(b-h)$. Because $0 \leq g_{b-h} - g_b \leq hH_o$, compactness and uniqueness of the old pole imply $x_h \rightarrow x$. The strict local quadratic growth in Proposition 2.8 gives, in normal coordinates $\Delta_h = \exp_x^{-1} x_h$,

$$|\Delta_h| = O(\sqrt{h}), \quad 0 \leq A(b-h) - A(b) \leq hH_o. \quad (4.26)$$

Choose one duration $q \in (t, 1)$ and split each active endpoint cost at $z_i = \exp_x(qp_i)$. The energy inequality

$$c(x', y_i) \leq c_q(x', z_i) + c_{1-q}(z_i, y_i)$$

is an equality at $x' = x$. Its first term is smooth near x , because qp_i is before cut. Its gradient is $-p_i$ and its Hessian is $A_x(qp_i)/q$. Combining this upper support with the smooth prefix cost gives

$$\begin{aligned} A(b-h) &\geq A(b) + hB_o(s_i) + \langle p_i - p, \Delta_h \rangle + \frac{1}{2}Q_i(\Delta_h, \Delta_h) + o(|\Delta_h|^2), \\ Q_i &= \frac{1}{t}A_x(tp) - \frac{1}{q}A_x(qp_i). \end{aligned} \quad (4.27)$$

All expansions here concern fixed scales, a fixed time, and a fixed finite representation. Equations (4.26)–(4.27) bound each $\langle p_i - p, \Delta_h \rangle$ above by $O(h)$. Their positive weighted sum is zero, so each is also bounded below by $-O(h)$. Hence, for $E = \text{span}\{p_i - p\}$,

$$|\text{proj}_E \Delta_h| = O(h). \quad (4.28)$$

On $E^\perp$, MTW concavity on the simplex of the qp_i gives

$$A_x(qp) \geq \sum_i m_i A_x(qp_i).$$

Radial monotonicity of divided action, equivalently Lemma 2.3, then gives

$$\sum_i m_i Q_i = \frac{1}{t}A_x(tp) - \sum_i \frac{m_i}{q}A_x(qp_i) \geq 0 \quad \text{on } E^\perp. \quad (4.29)$$

Average (4.27). The linear terms cancel. By (4.26), (4.28), and (4.29), its quadratic term is bounded below by $-o(h)$: the transverse part is nonnegative, mixed terms are $O(h^{3/2})$, and the remaining part is $O(h^2)$. The Taylor remainders are $o(h)$ as well. Thus

$$A(b-h) - A(b) \geq h \sum_i m_i B_o(s_i) + o(h).$$

The unique center minimizer and the elementary envelope argument (compare the old and new minimizers and use their convergence) give

$$C(b-h) - C(b) = -hB_c(s) + o(h).$$

Since b is an interior maximum of $A + C - \pi$, division of the resulting one-sided comparison by h and passage to the limit give (4.25). The argument applies to the arbitrary representation fixed at the start, proving the asserted universal quantifier. $\square$

Lemma 4.8 (Levels at a regularized maximum). *At a maximum in Lemma 4.6, the center and all active endpoints belong to $S_{(H_o+2)b}(x; u, v)$. For every active log p_i ,*

$$v(y_i) - v(y) = \frac{|p|^2 - |p_i|^2}{2} + b\{B_c(s) - B_o(s_i)\} - E_b(y). \quad (4.30)$$

In every positive barycentric representation the total weight on levels $s_i \geq 1 - \eta$ is at least c_1 . Moreover

$$s < 3/4, \quad \max\{s, s_i\} - \min\{s, s_i\} \leq 1 + \delta_j, \quad \min\{s, s_i\} \geq -2\eta \quad (4.31)$$

for small scales. These conclusions hold at each exact maximum, before taking a prefix-time limit.

Proof. An active endpoint has original gap $u - g_b - bB_o(s_i) \in [0, bH_o]$. By (4.24), the center gap is $u - g_b + E_b(y) - bB_c(s)$, which is at most $(H_o + 2)b$ by (4.19). This proves the common section assertion and its oscillation bound. Subtracting the active contact equality from (4.24) gives (4.30). Since $E_b(y) > 0$ and $E_b(y) \leq bB_c(s)$, one has $B_c(s) > 0$, and therefore $s < 3/4$.

Let m_R be the total right weight. Nonnegativity of B_o and its lower bound by one outside the right region give

$$1 - m_R \leq \sum_i m_i B_o(s_i) \leq B_c(s) - c_1 \leq 1 - c_1.$$

Thus $m_R \geq c_1$. In particular a right endpoint is present. The common range bound then puts every level above $(1 - \eta) - (1 + \delta_j) = -\eta - \delta_j \geq -2\eta$. $\square$

4.6 A quantitative outward endpoint

The next Lemma selects an endpoint once the active logs lie on a line and the center log has a nonzero component along that line. The center Hessian estimate in Section 5 will establish these two properties; the selection itself is finite-dimensional.

Lemma 4.9 (Weight, cap, and norm increment). *Suppose $p = \sum_{i=1}^N m_i p_i$ in an n -dimensional Euclidean space, with $N \leq n + 1$, $m_i > 0$, and $\sum_i m_i = 1$. Suppose $E = \text{span}\{p_i - p\} = \mathbb{R}e$, where $|e| = 1$, and $q = p \cdot e > 0$. Assume the template and level inequalities (4.25), (4.31), and*

$$\left| |p_i|^2 - |p|^2 + 2D(s_i - s) \right| \leq \zeta D, \quad 0 \leq \delta_j \leq \zeta \leq 1/16.$$

There is an outward index i with $p_i = p + d_i e$, $d_i > 0$, such that

$$m_i \geq m_* := \frac{c_1}{12(n+1)}, \quad m_i d_i \geq \frac{c_1 D}{8(n+1)q}, \quad d_i \leq \frac{3D}{2q}, \quad (4.32)$$

$$0 < |p_i|^2 - |p|^2 \leq 3D, \quad (4.33)$$

$$-2\eta \leq s_i < 1 - \eta, \quad B_o(s_i) \leq 1/m_*. \quad (4.34)$$

In particular the cap $M_0 = 2/m_$ can be fixed before K . All these constants are independent of K .*

Proof. Write $p_i = p + d_i e$. The level inequality gives total right mass at least c_1 , as in Lemma 4.8. A right point has $s_i - s > 1/4 - \eta$, hence

$$|p_i|^2 - |p|^2 = 2qd_i + d_i^2 \leq -D/4.$$

It follows that $d_i < 0$ and $|d_i| \geq D/(8q)$. Every outward point satisfies, by the common level range,

$$0 < 2qd_i + d_i^2 \leq 2D(1 + \delta_j) + \zeta D \leq 3D,$$

and so $d_i \leq 3D/(2q)$. The barycenter identity $\sum_i m_i d_i = 0$ therefore implies

$$\sum_{d_i > 0} m_i d_i \geq \frac{c_1 D}{8q}.$$

Choose an outward index contributing at least $1/N$ of this moment. Its upper displacement bound gives $m_i \geq c_1/(12N) \geq m_*$, proving (4.32).

The parameter inequality and nonnegativity of the outer template give $m_i B_o(s_i) \leq 1$, proving its cap bound. The lower level bound is already known. The positive norm increment implies $s_i \leq s + \zeta/2 < 3/4 + 1/32 < 1 - \eta$. Thus the selected endpoint lies in the region where (4.9) applies when $M_0 = 2/m_*$. $\square$

For a convergent sequence of active representations, an endpoint selected by Lemma 4.9 in the limit carries a definite positive moment as well as a definite weight. Its strict outward displacement and its strict separation from the right region persist for the corresponding approximating vertices. The remaining analytic task is to produce the one-dimensional difference space and a bound $q^2 \geq \kappa D$ with $\kappa > 0$ independent of K . Together with (4.33), these are the hypotheses of Corollary 3.6, which will compare the center and selected outer Jacobians.

5 Lower jets and the determinant contradiction

The comparison maxima of Section 4 relate a center endpoint to finitely many active outer endpoints. We seek determinant bounds for lower tests of the original potential at these endpoints. The main obstacle is that the center selected by a maximum need not belong to the full-measure set where the original transport Jacobian equation holds. Even though the short-time minimizing map is Lipschitz, it could send a set of positive volume into that exceptional set.

We first transfer lower jets from the prefix regularization to its active outer endpoints. The original contact-volume bound applies to these lower tests and bounds the selected outer determinant. That bound controls the regularized center Hessian, which in turn makes the short-time minimizing map nonsingular at the sampled points. Only then can we use the original Jacobian equation at the centers. The resulting matrix inequalities give a determinant gain proportional to the outer curvature parameter K , contradicting the geometric comparison from Section 3.

Throughout the argument the manifold and density bounds are fixed. In the local calculations, F, G and their indexed versions denote regularized potentials; the scalar oscillation function F from Section 4 returns in Proposition 5.10.

Throughout this section, a *lower Taylor jet* (ζ, B) of a function f at the origin means

$$f(h) \geq f(0) + \zeta \cdot h + \frac{1}{2}B(h, h) + o(|h|^2).$$

Replacing B by $B - \varepsilon \text{Id}$, for any $\varepsilon > 0$, gives a genuine quadratic lower test on a sufficiently small neighborhood. All the tests to which we apply a determinant inequality will have a strictly positive relative Hessian, so this replacement and the subsequent limit $\varepsilon \downarrow 0$ are harmless.

5.1 The original contact measure and lower tests

Lemma 5.1 (Contact-volume comparison). *Let u, v be the dual potentials for an admissible pair of densities, and let*

$$\mathcal{C} = \{(x, y) : u(x) + c(x, y) + v(y) = 0\}.$$

For a Borel set A on either side, its contact image satisfies

$$\frac{\lambda}{\Lambda} \text{vol}(A) \leq \text{vol}(\mathcal{C}(A)) \leq \frac{\Lambda}{\lambda} \text{vol}(A). \quad (5.1)$$

Projections are interpreted in the completed volume measures.

Proof. The contact relation is closed. Its projections of Borel sets are analytic, hence measurable for the completed measures. The optimal coupling has full mass on $\mathcal{C}$. At every differentiability point of either potential its contact endpoint is unique: all minimizing first-variation logs must equal its gradient, and their exponential endpoints therefore coincide. Since both marginals are absolutely continuous, almost every contact pair lies in the sets where both potentials are differentiable. The two contact maps on these sets are inverse to each other.

Write $\mu = \rho_0 \text{vol}$ and $\nu = \rho_1 \text{vol}$. Consequently $\nu(\mathcal{C}(A)) = \mu(A)$ for sets on the first side, and $\mu(\mathcal{C}(A)) = \nu(A)$ for sets on the second side: any additional contact image outside the almost-everywhere map image consists, up to a null set, of points where the opposite potential is not differentiable. The density bounds give (5.1). $\square$

Lemma 5.2 (Upper determinant bound for a lower test). *Let v be one of the original dual potentials. Suppose a C^2 function ϕ touches v from below at y , its gradient $r = \text{grad } \phi(y)$ belongs to $I(y)$, and*

$$W = \text{Hess } \phi(y) + A_y(r) > 0.$$

Then

$$\sigma_y(r) \det W \leq \Lambda/\lambda. \quad (5.2)$$

The same conclusion holds for a lower Taylor jet with these properties, by approximation of its quadratic part.

Proof. It suffices first to prove the result after decreasing the Hessian by a small positive multiple of the metric. Thus we may arrange strict touching, with $v - \phi \geq \varepsilon d(\cdot, y)^2$ in a small coordinate ball. The map

$$S(z) = \exp_z \text{grad } \phi(z)$$

is a local diffeomorphism near y . Indeed its defining equation is $D_z c(z, S(z)) + \text{grad } \phi(z) = 0$, and differentiation gives

$$|\det DS(z)| = \sigma_z(\text{grad } \phi(z)) \det\{\text{Hess } \phi(z) + A_z(\text{grad } \phi(z))\}. \quad (5.3)$$

The relative Hessian is positive throughout a sufficiently small ball, and the logs remain before cut.

For $h > 0$, let Ω_h be the connected component containing y of $\{\phi + h > v\}$ inside that ball. It is relatively compact and shrinks to y . If $z \in \Omega_h$ and $x = S(z)$, the cost mountain

$$m_x(w) = \phi(z) + c(z, x) - c(w, x)$$

is tangent to ϕ at z . On a fixed smaller coordinate ball, $\phi - m_x$ has positive Hessian, uniformly for x close to $S(y)$; thus $\phi \geq m_x$ there. On $\partial\Omega_h$ we have $v - m_x \geq h$, whereas at z we have $v(z) - m_x(z) < h$. The minimum of $v - m_x$ on $\overline{\Omega_h}$ is therefore attained at an interior point z' . Thus $\text{grad } m_x(z') \in \partial v(z')$. By Theorem 2.5, the corresponding local contact is a global cost contact. Its endpoint is x . This proves

$$S(\Omega_h) \subset \mathcal{C}(\Omega_h).$$

Use the area formula for the diffeomorphism S [15, Chapter 2, Theorem 3.3], followed by Lemma 5.1, to obtain

$$\inf_{\Omega_h} |\det DS| \text{vol}(\Omega_h) \leq \text{vol}(S(\Omega_h)) \leq (\Lambda/\lambda) \text{vol}(\Omega_h).$$

Let $h \downarrow 0$ and then restore the original Hessian. $\square$

Only the original correspondence occurs in these two Lemmas. No density bound is claimed for a modified potential or its envelope. For later use, the almost-everywhere counterpart of (5.3) is the following consequence of [2, Proposition 4.1 and Theorem 4.2]. There is a volume-null Borel set $N \subset M$ such that, at every $y \notin N$, v has an Alexandrov Hessian, its true $\log r = \text{grad } v(y)$ is before cut, and

$$W_v(y) := \text{Hess } v(y) + A_y(r) > 0, \quad \frac{\lambda}{\Lambda} \leq \sigma_y(r) \det W_v(y) \leq \frac{\Lambda}{\lambda}. \quad (5.4)$$

We enlarge N to include the exceptional sets in the contact-map differential of [2, Proposition 4.1(b)]. The Jacobian equation in [2, Theorem 4.2] assumes only absolute continuity with L^1 densities. The lower density bound makes its full-source-measure set a full-volume set; differentiability of a density is not required.

5.2 An exact endpoint-jet calculation

Fix a continuous datum w_o and put

$$g(x) = \max_y \{-c(x, y) - w_o(y)\}, \quad F = Q_t g, \quad 0 < t < 1.$$

By Proposition 2.8, F is locally $C^{1,1}$ and its minimizing pole $x(z)$ is unique and locally Lipschitz. The following calculation concerns the first minimum alone.

Lemma 5.3 (The pole matrix and endpoint lower jets). *Suppose z_0 is a point where F has a second-order expansion and $x(z)$ is differentiable. Write $x = x(z_0)$, $z_0 = \exp_x(tp)$, and choose active logs*

$$p = \sum_{j=1}^N m_j p_j, \quad m_j > 0, \quad \sum_j m_j = 1, \quad N \leq n+1,$$

where $y_j = \exp_x p_j$ and $g(x) + c(x, y_j) + w_o(y_j) = 0$. Set $E = \text{span}\{p_j - p : 1 \leq j \leq N\}$. In the fixed-pole coordinates $z(q) = \exp_x(tq)$, define

$$L = D_q^2 \{c_t(x, z(q)) - F(z(q))\}|_{q=p}. \quad (5.5)$$

Then

$$L \geq 0, \quad LE = 0. \quad (5.6)$$

For every j , the function

$$\Psi_j(d) = w_o(\exp_x(p_j + d)) + \frac{1}{2}|p_j + d|^2$$

has at $d = 0$ the lower Taylor jet $(0, m_j L)$, after subtracting its value at zero. If p_j is nonconjugate, this is an endpoint lower test in the smooth chosen cost branch.

More precisely, for $t < s < 1$ put

$$Q_s = \frac{1}{t} A_x(tp) - \frac{1}{s} \sum_j m_j A_x(sp_j).$$

Then $Q_s \geq 0$ on $E^\perp$, and $m_j(L + LQ_s L)$ is a lower Taylor Hessian for Ψ_j .

Proof. Use normal coordinates $x' = \exp_x a$ and set

$$B(a, q) = c_t(\exp_x a, z(q)), \quad f(q) = F(z(q)), \quad a(q) = \log_x x(z(q)).$$

At the reference point $a(p) = 0$. First variation gives the exact identities

$$B(0, q) = \frac{t}{2}|q|^2, \quad B_a(0, q) = -q.$$

Thus $B_{qq} = t\text{Id}$, $B_{qa} = -\text{Id}$ at $(0, p)$. The envelope identity $Df(q) = B_q(a(q), q)$ and differentiability of the pole imply

$$D^2f = t\text{Id} - R, \quad R = D_q a(p) = L. \quad (5.7)$$

In particular R is symmetric. Since $f(q) \leq g(x) + B(0, q)$ with equality at p , we have $L \geq 0$.

Choose a common $s \in (t, 1)$. For each j define

$$y_j(d) = \exp_x(p_j + d), \quad z_j(d) = \exp_x(s(p_j + d)),$$

and the split upper cost

$$C_j(a, d) = \frac{c(\exp_x a, z_j(d))}{s} + \frac{c(z_j(d), y_j(d))}{1-s}. \quad (5.8)$$

Both cost pieces are smooth near $(0, 0)$ because they are proper subsegments of a minimizing ray. Energy splitting gives $c(\exp_x a, y_j(d)) \leq C_j(a, d)$. At $a = 0$, the chosen geodesic remains minimizing on both pieces for small d , and consequently

$$C_j(0, d) = \frac{1}{2}|p_j + d|^2, \quad (C_j)_a(0, d) = -(p_j + d), \quad (C_j)_{ad} = -\text{Id}, \quad (C_j)_{aa} = \frac{1}{s}A_x(sp_j). \quad (5.9)$$

The use of the fixed reference pole in $z_j(d)$ is what makes the mixed derivative exactly $-\text{Id}$.

The nonnegative function

$$g(x(z(q))) + C_j(a(q), 0) + w_o(y_j)$$

vanishes at p . It is differentiable there, since $g(x(z(q))) = f(q) - B(a(q), q)$, and its derivative equals $(p - p_j)^* R$. This derivative is zero. Symmetry in (5.7) proves $LE = 0$.

On $E^\perp$, MTW concavity on the convex hull of the sp_j , followed by divided-action monotonicity, gives

$$Q_s \geq \frac{1}{t}A_x(tp) - \frac{1}{s}A_x(sp) > 0.$$

All points used in this inequality are before cut by Theorem 2.5; equivalently one may invoke Lemma 2.3. Notice that unsplit endpoints may still be conjugate.

Fix i and write $m = m_i$. Given d , choose any linear map $U : T_x M \rightarrow E$, and set $k = m(d + Ud)$ and $a = a(p + k)$. In the lower bound defining $g(\exp_x a)$, move endpoint i to $y_i(d)$, keep all other endpoints fixed, replace the costs by (5.8), and average with the old weights. Write $\hat{\Psi}_i(d) = \Psi_i(d) - \Psi_i(0)$. The resulting inequality is

$$f(p + k) \geq B(a, p + k) - \sum_j m_j \{C_j(a, d_j) + w_o(y_j(d_j))\},$$

where $d_i = d$ and $d_j = 0$ for $j \neq i$. Subtract the equality at $(a, k, d) = (0, 0, 0)$ and use (5.7)–(5.9). The linear terms cancel: the pole terms cancel because $\sum m_j p_j = p$, and the prefix terms cancel because $Df(p) = tp$. The remaining inequality is

$$m\hat{\Psi}_i(d) \geq \frac{1}{2}L(k, k) + \frac{1}{2}Q_s(a, a) + a \cdot (md - k) + o(|d|^2). \quad (5.10)$$

Here $a = Lk + o(|d|)$. Since $md - k = -mUd$ and $\text{range } L \subset E^\perp$, the mixed term is $o(|d|^2)$. Hence the lower Hessian is

$$m(\text{Id} + U)^*(L + LQ_sL)(\text{Id} + U).$$

In fact this equals $m(L + LQ_sL)$ because $LU = 0$. The latter dominates mL , proving the asserted jet. The linear term is exactly zero independently of any second function or maximum condition.

The map $d \mapsto y_i(d)$ need not be invertible for the preceding pullback calculation. When p_i is nonconjugate it is a local diffeomorphism, and $\frac{1}{2}|p_i + d|^2$ is the smooth chosen upper branch of $c(x, y_i(d))$. This converts the pullback jet into an endpoint jet. $\square$

Let G be another locally $C^{1,1}$ function, and define

$$H = D_q^2\{c_t(x, z(q)) + G(z(q))\}|_{q=p}. \quad (5.11)$$

The exact identity

$$H - L = D_q^2\{(F + G)(z(q))\}|_{q=p} \quad (5.12)$$

will distinguish the first-minimum calculation from the maximum argument. At a twice differentiable maximum, it gives $H \leq L$ and $H|_E \leq 0$. At a Jensen point with gradient and upper Hessian errors tending to zero in a fixed smooth chart, it gives

$$H \leq L + \eta \text{Id}, \quad \eta \longrightarrow 0. \quad (5.13)$$

The coordinate-change term uses the gradient error and also tends to zero. Lemma 5.3 is exact at such a point. For every linear U with image in E , it therefore supplies a lower jet $H_i = m_iL$ satisfying

$$H_i \geq m_i(\text{Id} + U)^*H(\text{Id} + U) - m_i\eta(\text{Id} + U)^*(\text{Id} + U).$$

In particular no approximate stationarity term occurs in the linear coefficient of the endpoint test.

5.3 The center semibound without a density equation

We use the scales and templates of the preceding section. To distinguish a scalar argument from a point of the manifold, write

$$\varphi_c(h) = h + bB_c((h - a)/D), \quad \varphi_o(h) = h + bB_o((h - a)/D), \quad w_c = \varphi_c \circ v, \quad w_o = \varphi_o \circ v.$$

For each fixed outer template the scales are sufficiently small that both scalar maps have derivative between 1/2 and 2. On the fixed compact interval of possible center levels, B'_c is negative and bounded away from zero. Its derivative bounds are fixed independently of the outer curvature parameter K .

Lemma 5.4 (A lower semibound before the center Jacobian). *Fix one set of scales, one original potential v , and the two templates. Let $t_j \uparrow 1$ and $b_j \in [b_-, b_+]$. Put*

$$F_j = Q_{t_j}g_{b_j}, \quad G_j = Q_{1-t_j}w_{c,b_j}.$$

Let z_j be simultaneous differentiability points for the objects in Lemma 5.3, with second-order expansions of G_j , whose center levels lie in the fixed compact interval above. Suppose

$$|\text{grad}(F_j + G_j)(z_j)| \longrightarrow 0.$$

Let x_j be the prefix pole and $z_j = \exp_{x_j}(t_j p_j)$. The matrix H_j of (5.11) satisfies

$$H_j \geq c \frac{b_j}{D} \text{Id} - C \frac{b_j}{D^2} p_j p_j^* - \varepsilon_j \text{Id}, \quad \varepsilon_j \longrightarrow 0. \quad (5.14)$$

The constants c, C depend on the geometry and the fixed center template, but not on K . The errors tend to zero with the prefix and gradient errors at the fixed scales. No upper bound on H_j , and no Alexandrov Hessian of v at the center, is assumed.

Proof. Write $\tau_j = 1 - t_j$, and let y_j be the unique final minimizer. Set

$$r_j = \tau_j^{-1} \log_{y_j} z_j, \quad l_j = \varphi'_c(v(y_j)) < 1.$$

The short minimum gives a smooth local lower test for w_c with gradient r_j . Inverting φ_c gives a local lower test for v with gradient r_j/l_j . Theorem 2.5 therefore implies that

$$\zeta_j = \exp_{y_j}(r_j/l_j)$$

is a true contact endpoint of v , joined by the indicated minimizing ray. In particular $\hat{x}_j = \exp_{y_j} r_j$ is strictly before cut along that ray.

For the moment omit subscripts. Put $\theta = (1 + l^{-1})/2$ and $\xi = \exp_y(\theta r)$. Split the true support at ξ , using the fraction $l\theta$ of its unit duration. The resulting upper cost support is

$$C(h) = \frac{c(h, \xi)}{l\theta} + \frac{c(\xi, \zeta)}{1 - l\theta}.$$

It agrees with $c(h, \zeta)$ at $h = y$, and $v(h) \geq v(y) + c(y, \zeta) - C(h)$. Thus

$$\psi(h) = \varphi_c\{v(y) + c(y, \zeta) - C(h)\}$$

is a smooth local lower support for w_c , with

$$\text{grad } \psi(y) = r, \quad \text{Hess } \psi(y) = -\theta^{-1} A_y(\theta r) + \frac{\varphi''_c(v(y))}{l^2} r r^*. \quad (5.15)$$

The original true endpoint ζ may be conjugate. The point ξ is halfway into the available extension and is before cut, which is all that this construction uses.

At the fixed scales, $1 - l$ has a positive lower bound and the parameters $(y, r, l, v(y))$ range over a compact set. The first cost piece in C is uniformly smooth on a fixed neighborhood of y ; the second piece is constant in h . Consequently the supports ψ have uniform smooth bounds and a uniform supporting radius. For sufficiently small τ , the local infimum $Q_\tau \psi$ is smooth near z , has minimizer y at z , and supports $Q_\tau w_c$ from below there. To justify the last assertion, all minimizers of the latter lie within $O(\tau)$ of z , by the Lipschitz bound on w_c , and so lie in the supporting neighborhood. For ψ the local Hessian of the minimizing expression is $\tau^{-1} \text{Id} + O(1) > 0$. Equality and stationary first variation hold at y . The implicit-function theorem and the Schur formula for a minimum give, after parallel identification,

$$\text{Hess } Q_\tau \psi(z) = \text{Hess } \psi(y) + O(\tau). \quad (5.16)$$

All constants in this statement are uniform for the fixed-scale family just described.

Let $\hat{p} = \log_{\hat{x}} y$ and $\hat{L} = (d \exp_{\hat{x}})_{\hat{p}}$. Lemma 3.4, applied to the minimizing ray through duration $1/l$, gives

$$\hat{L}^* \{A_y(r) - \theta^{-1} A_y(\theta r)\} \hat{L} \geq c(1 - l) \text{Id}.$$

Gauss's Lemma gives $\widehat{L}^*r = -\widehat{p}$. Combining this with (5.15) yields

$$\widehat{L}^*\{\text{Hess } \psi(y) + A_y(r)\}\widehat{L} \geq c(1-l)\text{Id} + \frac{\varphi_c''(v(y))}{l^2}\widehat{p}\widehat{p}^*. \quad (5.17)$$

Here $1-l \geq cb/D$ and $|\varphi_c''(v(y))|/l^2 \leq Cb/D^2$.

We finally compare the reference pole with $\widehat{x}_j$. The envelope identities give

$$x_j = \exp_{z_j}(-t_j \text{grad } F_j(z_j)), \quad \widehat{x}_j = \exp_{z_j}(t_j \text{grad } G_j(z_j)).$$

They are at distance tending to zero. The pairs $(\widehat{x}_j, y_j)$ have a uniform fixed-scale extension, so their cost charts and inverse exponential charts are uniformly smooth. Omit subscripts again and put $U = Q_\tau \psi$, $Z(q) = \exp_x(tq)$, $M = D_q Z(p)$, and $R = \nabla_q^2 Z(p)$. Since $G - U$ has a local minimum at z , its first derivative vanishes and its second-order form is positive semidefinite. Consequently

$$H \geq H_\psi := t\text{Id} + M^* \text{Hess } U(z)M + \langle \text{grad } U(z), R \rangle.$$

Only the smooth support U occurs on the right. Its gradient and Hessian are uniformly bounded at the fixed scales, and (5.16) identifies its Hessian with that of ψ , up to $O(\tau)$ after parallel transport.

Write $\delta = |\text{grad}(F+G)(z)|$ and identify nearby pole tangent spaces by parallel transport. The explicit pole formulas give $d(x, \widehat{x}) = O(\delta)$. Moreover $p = t^{-1} \log_x z$ and $\widehat{p} = t^{-1} \log_{\widehat{x}} z$ differ by $O_{\text{fixed}}(\delta)$, since the latter ray has a uniform fixed-scale extension. Thus replacing x, p by $\widehat{x}, \widehat{p}$ in M, R changes their entries by $O_{\text{fixed}}(\delta)$. Changing t to 1 then changes them by $O(\tau)$. Every resulting error in H_ψ multiplies a bounded derivative of U .

For the full endpoint map $Y(q) = \exp_{\widehat{x}} q$, let $B = DY(\widehat{p})$. The identity $c(\widehat{x}, Y(q)) = |q|^2/2$ on the chosen branch gives

$$\text{Id} = B^* A_y(r)B - \langle r, \nabla_q^2 Y(\widehat{p}) \rangle.$$

Since $\text{grad } \psi(y) = r$, the preceding expansions yield

$$H \geq H_\psi = B^*\{\text{Hess } \psi(y) + A_y(r)\}B + O_{\text{fixed}}(\tau + \delta).$$

Combining this with (5.17) proves (5.14). No coordinate error multiplies the unknown Hessian of G . $\square$

The true and modified endpoints occur in opposite orders along the center and outer reverse rays. In the center calculation, $l < 1$ makes $\widehat{x} = \exp_y r$ a proper intermediate point on the true ray to $\zeta = \exp_y(r/l)$. At an active outer endpoint y_i , a slope $l_i = \varphi'_o(v(y_i)) > 1$ instead makes the true endpoint $\zeta_i = \exp_{y_i}(r_i/l_i)$ an intermediate point on the modified ray to $x = \exp_{y_i} r_i$, where r_i is its minimizing reverse log. Figure 1 displays this distinction. We next use it to exclude conjugacy at the full modified outer endpoint.

5.4 The outward endpoint and conjugacy

Lemma 5.5 (Conjugacy is excluded by an active increasing modification). *Fix $b > 0$ and an active outer pair*

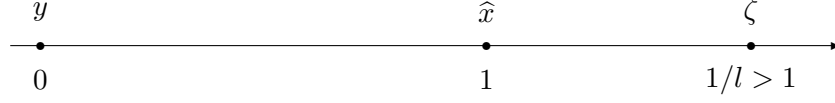
$$g_b(x) + c(x, y) + \varphi_o(v(y)) = 0.$$

Choose any minimizing reverse log r from y to x . Suppose

$$l = \varphi'_o(v(y)) > 1, \quad \varphi''_o(v(y)) < 0.$$

Then the chosen modified ray is nonconjugate at its endpoint. The true log r/l is before cut.

Center modification: $l < 1$



Selected outer modification: $l_i > 1$

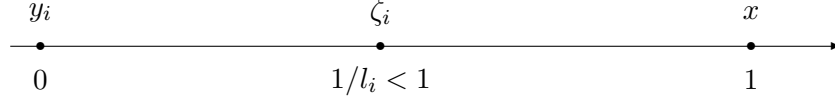


Figure 1: Two independent reverse rays, shown schematically with their duration parameters. The modified center endpoint $\hat{x}$ and the true outer endpoint ζ_i are strictly before cut. The full endpoints ζ and x may still be conjugate at this stage of the argument. The two rows are not drawn to a common scale.

Proof. The active inequality reads $\varphi_o(v(h)) \geq -c(x, h) - g_b(x)$. For $1/l < s < 1$, put $\xi_s = \exp_y(sr)$ and form the smooth upper support

$$C_s(h) = c(h, \xi_s)/s + c(\xi_s, x)/(1 - s).$$

It has value $c(y, x)$ and gradient $-r$ at y . Since φ_o is increasing, $\varphi_o^{-1}(-C_s - g_b(x))$ is a smooth lower test for v with gradient r/l . That log is strictly before cut because it shortens a minimizing ray. It also gives a global contact by Theorem 2.5, but its before-cut property does not depend on a density equation.

The scaled true relative Hessian of this test is

$$W_s = lA_y(r/l) - s^{-1}A_y(sr) - \frac{\varphi_o''(v(y))}{l^2}rr^*. \quad (5.18)$$

Fix $s_0 \in (1/l, 1)$. For $s \geq s_0$, divided-action monotonicity gives a positive definite lower bound for the difference of the first two terms, independent of s . The last term is positive semidefinite. If the endpoint at duration one were conjugate, the index-form characterization of conjugacy, equivalently Lemma 3.2, would force an eigenvalue of $s^{-1}A_y(sr)$ to tend to $-\infty$ as $s \uparrow 1$. Thus $\det W_s \rightarrow \infty$, while W_s retains a positive definite baseline. Lemma 5.2, at the fixed true log r/l , instead gives

$$\det W_s \leq l^n(\Lambda/\lambda)/\sigma_y(r/l) < \infty.$$

This is a contradiction. □

Lemma 5.6 (Undoing the outer modification). *At a pair as in Lemma 5.5, let $p = \log_x y$ be the corresponding forward log, $B = (d \exp_x)_p$, and let $H_o \geq 0$ be a lower Taylor Hessian, with zero linear part, of $d \mapsto w_o(\exp_x(p + d)) + \frac{1}{2}|p + d|^2$. Define*

$$V_o = H_o + B^*\{lA_y(r/l) - A_y(r)\}B - \frac{\varphi_o''(v(y))}{l^2}pp^*. \quad (5.19)$$

Then $V_o > 0$ and

$$\det V_o \leq l^n(\Lambda/\lambda) \frac{\sigma_x(p)^2}{\sigma_y(r/l)}. \quad (5.20)$$

The middle term in (5.19) is positive definite. On a compact family of active pairs where $b > 0$ is bounded away from zero, $\varphi'_o > 1$ and $\varphi''_o < 0$ hold with fixed strict margins, this term has a uniform positive definite baseline and the chosen exponential branches are uniformly nonsingular.

Proof. Lemma 5.5 makes the endpoint coordinate map locally invertible. Its smooth cost branch has endpoint Hessian $A_y(r)$, and the zero linear coefficient fixes the gradient of the induced lower test of w_o to be r . Inverting φ_o and using the chain rule shows that (5.19) is l times the true relative lower-test Hessian, pulled back by B . Gauss's Lemma gives the rank-one term displayed there.

The divided-action difference between durations $1/l$ and 1 is strictly positive definite, so $V_o > 0$. Decrease the lower Taylor Hessian by εId to obtain a genuine lower test, apply Lemma 5.2, and let $\varepsilon \downarrow 0$. Taking the determinant of the scaling and pullback gives (5.20).

For the uniformity assertion, include the minimizing log among the parameters. The family is compact and active contact is a closed condition. Lemma 5.5 excludes a conjugate limit anywhere in that family; radial shortening excludes true cut limits. The smooth branch matrices and the positive divided-action difference therefore have the claimed uniform bounds by compactness. Multiple ordinary branches cause no difficulty: each specified nonconjugate minimizing branch is a smooth upper branch and has been included among the parameters. $\square$

5.5 The order of the limiting argument

We now select points at which the preceding lower tests and the original Jacobian equation can be combined. The following elementary fact will justify differentiation through a short-time minimizer once its endpoint is an Alexandrov point: neighboring endpoints need not be differentiable points of the datum.

Lemma 5.7 (Subgradients at an Alexandrov point). *If f is semiconvex near 0 and has the Alexandrov expansion*

$$f(h) = f(0) + p \cdot h + \frac{1}{2}A(h, h) + o(|h|^2),$$

then every nearby subgradient satisfies

$$\zeta = p + Ah + o(|h|), \quad \zeta \in \partial f(h),$$

uniformly over the choice of ζ .

Proof. Adding a fixed quadratic reduces the assertion to a convex function, and subtracting it later restores the displayed formula. In the convex case write the remainder as $R(h)$, where $|R(h)| \leq \omega(|h|)|h|^2$ for a nondecreasing modulus $\omega(r) \rightarrow 0$. For $r = |h| > 0$, a unit vector e , and $0 < \alpha \leq 1$, the subgradient inequalities at $h \pm \alpha re$ give

$$|(\zeta - p - Ah) \cdot e| \leq \frac{1}{2}\|A\|\alpha r + 5\omega(2r)r/\alpha.$$

For small r , choose $\alpha = \sqrt{\max\{\omega(2r), r^2\}}$ and take the supremum over unit e . This proves the uniform assertion. $\square$

The selection has two stages. At each fixed prefix time we first pass from nearby twice differentiable points to an exact maximum, where the parameter inequality holds for every active representation. After selecting an outer endpoint in that limit, we return to the nearby points to obtain the Hessian bound and the original center Jacobian. This order preserves the parameter inequality without assuming that a prescribed active vertex persists under perturbation.

Proposition 5.8 (The limiting matrix configuration). *Fix the center template and the constants of Lemmas 4.5, 4.6, 4.7, and 4.9. Choose K , then an outer template with curvature less than $-K$ in the prescribed capped region. For all sufficiently small scales in the sequence of Lemma 4.2, there are a parameter $b \in [b_-, b_+]$, a pole x , center and active logs $p, p_i \in G_x$, weights $m_i > 0$ summing to one, and symmetric matrices H, L, V, V_i on $T_x M$, with the following properties. For one selected index i ,*

$$p = \sum_j m_j p_j, \quad E = \text{span}\{p_j - p\} = \mathbb{R}e, \quad |e| = 1, \quad q = p \cdot e > 0, \quad q^2 \geq cD, \quad (5.21)$$

$$m_i \geq m_* > 0, \quad p_i = p + d_i e, \quad d_i > 0, \quad (5.22)$$

$$L \geq 0, \quad LE = 0, \quad H \leq L, \quad H \geq V + c \frac{b}{D} \text{Id} - C \frac{b}{D^2} pp^*, \quad V > 0, \quad (5.23)$$

$$V_i \geq m_i L + cK \frac{b}{D^2} p_i p_i^*, \quad V_i > 0. \quad (5.24)$$

The center modified log and the selected outer modified log are nonconjugate. Let r and r_i denote the reverse logs of these specified rays, let $y = \exp_x p$, $y_i = \exp_x p_i$, and set $l = \varphi'_c(v(y))$, $l_i = \varphi'_o(v(y_i))$. The chosen true limiting branches are also nonconjugate, and

$$\det V \geq c \frac{\sigma_x(p)^2}{\sigma_y(r/l)}, \quad \det V_i \leq C \frac{\sigma_x(p_i)^2}{\sigma_{y_i}(r_i/l_i)}. \quad (5.25)$$

Here c, C, m_* are independent of K and the outer template. The level and norm-increment conclusions of Lemma 4.9 hold as well.

Proof. Step 1: fixed scales and exact maxima. Throughout the proof the original potential v , the scales, and both templates are fixed. Constants used only to pass limits may depend on all of them. Choose $t_k \uparrow 1$ and interior maxima (z_k^0, b_k) supplied by Lemma 4.6, for

$$F_k(z) = Q_{t_k} g_{b_k}(z), \quad G_k(z) = Q_{1-t_k} w_{c, b_k}(z).$$

The functions are $C^{1,1}$ in space: Proposition 2.8 applies to the first, and Lemma 2.4 to the second. At an exact maximum the velocities align. If x_k^0 and y_k^0 are its two minimizers and $p_k^0 = t_k^{-1} \log_{x_k^0} z_k^0$, then $y_k^0 = \exp_{x_k^0} p_k^0$ along the concatenated minimizing ray. Every finite positive active representation of p_k^0 satisfies the parameter inequality of Lemma 4.7. The level relations and common section bounds of Lemma 4.8 hold for these representations.

Step 2: null sets are removed before Jensen sampling. Let N be the null set for the original potential in (5.4). For this fixed k , the final minimizer map $Y_k(z)$ is locally Lipschitz and hence differentiable almost everywhere [15, Chapter 2, Theorem 1.4]. In each coordinate chart the locally Lipschitz area formula [15, Chapter 2, Theorem 3.3] implies that

$$Z_k = \{z : DY_k(z) \text{ exists, } \det DY_k(z) \neq 0, Y_k(z) \in N\} \quad (5.26)$$

is a null set. Indeed the integral of $|\det DY_k|$ on $Y_k^{-1}(N)$ is zero. This argument does not require us to know in advance where DY_k is nonsingular.

Apply the semiconvex maximum Lemma [3, Appendix, Lemmas A.3–A.4] to $F_k + G_k$ at z_k^0 , using a small fourth-order strictness penalty if needed. Its positive-measure conclusion allows us to choose a sequence $z_{k,\nu} \rightarrow z_k^0$ off Z_k and off all the null sets where F_k, G_k , the prefix pole map, or Y_k lack the derivatives used below. We have

$$\text{grad}(F_k + G_k)(z_{k,\nu}) \longrightarrow 0, \quad D^2(F_k + G_k)(z_{k,\nu}) \leq \eta_{k,\nu} \text{Id}, \quad \eta_{k,\nu} \longrightarrow 0 \quad (5.27)$$

in the fixed chart. At every such point choose a representation with at most $n + 1$ active logs, padding by zero weights only for compactness. Lemma 5.3 applies at each point. Its matrices $L_{k,\nu}$ are positive semidefinite and annihilate the active differences, exactly; its endpoint lower jets are $m_{i,k,\nu}L_{k,\nu}$. The matrices $H_{k,\nu}$ satisfy (5.13).

Step 3: take the first, fixed-prefix limits. For each fixed k , the local $C^{1,1}$ bounds are finite. After taking a subsequence in ν , all configuration parameters, weights, and the two matrices $H_{k,\nu}, L_{k,\nu}$ converge. Denote the limits by $p_k, p_{i,k}, m_{i,k}, H_k, L_k$ and the corresponding base points. Discard zero limiting weights. The limiting endpoints are active at the exact maximum, and their weighted sum is p_k . In particular the *universal* parameter inequality applies to this limiting representation; we have not asserted forward persistence of any prescribed active vertex. We retain

$$L_k \geq 0, \quad L_k E_k = 0, \quad H_k \leq L_k, \quad E_k = \text{span}\{p_{i,k} - p_k : m_{i,k} > 0\}. \quad (5.28)$$

The quantitative proof of Lemma 5.4 has an error bounded by a fixed-scale constant times $1 - t_k + |\text{grad}(F_k + G_k)(z_{k,\nu})|$. Passing $\nu \rightarrow \infty$ therefore gives

$$H_k \geq c \frac{b_k}{D} \text{Id} - C \frac{b_k}{D^2} p_k p_k^* - C_{\text{fixed}}(1 - t_k) \text{Id}.$$

Since $b_k \geq b_- > 0$, for sufficiently large k this implies, after reducing c ,

$$H_k \geq c \frac{b_k}{D} \text{Id} - C \frac{b_k}{D^2} p_k p_k^*. \quad (5.29)$$

If $\dim E_k \geq 2$, a nonzero vector in $E_k \cap p_k^\perp$ contradicts (5.28) and (5.29). If $E_k = 0$, all active logs in the representation equal p_k and all endpoints equal the center. But $B_c \leq B_o$ at every level, contradicting the strict parameter inequality. Thus $E_k = \mathbb{R}e_k$. Orient e_k so $q_k = p_k \cdot e_k > 0$; the same two inequalities give $q_k^2 \geq cD$. Lemma 4.9 now supplies an index with $m_{i,k} \geq m_*$, a strict outward displacement, and a bounded template value. We choose an index carrying a fixed share of the outward moment, as permitted by that Lemma, so its displacement has a positive lower bound at the fixed scales. Relabel it as index 1.

Step 4: fix the outer branch before bounding any center Hessian. Pass to a subsequence in k so that the selected configurations and their weights converge. Their centers and selected endpoints remain in the compact level ranges of Lemma 4.9. The selected limiting weight is at least m_* , its displacement is strictly outward, and its outer template value is below the prescribed cap. By imposing the template conditions under a slightly larger fixed cap, the limiting slope and curvature conditions have strict margins: $B'_o > 0$ and $B''_o < -K$. The limiting selected pair is active. Lemma 5.5 excludes conjugacy of its modified branch, using only this activity and the original contact-volume bound. Its shortened true log is before cut. Lemma 5.6 consequently supplies a neighborhood of this configuration with constants $\delta > 0$ and $M < \infty$ such that its divided-action gain is at least δId and its determinant upper bound is at most M . These constants may depend on the fixed scales and templates.

For every remaining k , return to its already chosen Jensen subsequence and choose $\nu = \nu(k)$ sufficiently large. We require the configuration and matrices to be within $1/k$ of their fixed-prefix limits, the selected weight to be at least $m_*/2$, and the selected endpoint to lie in that neighborhood. We also require both errors in (5.27), including their transformed versions in (5.13), to be at most $(1 - t_k)^3/k$. All these requirements are compatible because the limit in ν is taken with k fixed. For the rest of the proof, a subscript k denotes this diagonal sequence of actual Jensen points.

Step 5: the outer test bounds the exact pole matrix. At each diagonal point, Lemma 5.3 gives the actual lower Taylor Hessian $H_{o,k} = m_{1,k}L_k \geq 0$ at the selected endpoint. Use Lemma 5.6 to

undo the modification and call the resulting form $V_{1,k}$. The positive outer curvature term and the uniform divided-action baseline give

$$V_{1,k} \geq m_{1,k}L_k + \delta \text{Id}, \quad \det V_{1,k} \leq M. \quad (5.30)$$

Every eigenvalue of $V_{1,k}$ is at least δ , so its largest eigenvalue is at most $M\delta^{1-n}$. Since $m_{1,k} \geq m_*/2$, this bounds L_k from above. The comparison $H_k \leq L_k + o(1)\text{Id}$ and Lemma 5.4 now bound H_k from both sides. There has still been no use of a center density equation.

The center extension gives uniformly nonsingular fixed-scale prefix-log coordinate maps, as in the proof of Lemma 5.4. Their derivatives and second derivatives are bounded; the gradients of G_k are bounded by the Lipschitz bound of the datum. Thus the coordinate change in (5.11) converts the bound on H_k into

$$\|\text{Hess } G_k(z_k)\| \leq C_{\text{fixed}}. \quad (5.31)$$

Step 6: the chosen centers are good original endpoints. Write $\tau_k = 1 - t_k$, $y_k = Y_k(z_k)$, and $K_k(z, y) = c(z, y)/\tau_k$. The envelope identity $\text{grad } G_k = (K_k)_z(z, Y_k(z))$ and differentiability of Y_k give

$$\text{Hess } G_k = (K_k)_{zz} + (K_k)_{zy}DY_k.$$

In parallel orthonormal frames, the distance between z_k and y_k is $O(\tau_k)$, and the smooth short-cost expansions are

$$(K_k)_{zz} = \tau_k^{-1}\text{Id} + O(\tau_k), \quad (K_k)_{yy} = \tau_k^{-1}\text{Id} + O(\tau_k), \quad (K_k)_{zy} = -\tau_k^{-1}\Pi + O(\tau_k),$$

where Π is the appropriate parallel isometry. Solving the preceding identity, using (5.31), gives

$$DY_k = \Pi^{-1}\{\text{Id} - \tau_k \text{Hess } G_k\} + O(\tau_k^2). \quad (5.32)$$

It is nonsingular for large k . The Jensen points were chosen off the set (5.26); hence $y_k \notin N$. This is the promised justification for using the original Alexandrov and Jacobian statements at the centers.

In the following formulas the scalar modifications are evaluated with $b = b_k$. At these points $w_c = \varphi_c \circ v$ has an Alexandrov Hessian. The minimizing property supplies a proximal subgradient

$$\xi(z) = -(K_k)_y(z, Y_k(z)) \in \partial w_c(Y_k(z))$$

at every nearby point, even if w_c is not differentiable there. Apply Lemma 5.7 at the Alexandrov point $y_k = Y_k(z_k)$. Since Y_k is Lipschitz and differentiable at z_k , it gives, in normal coordinates at z_k and y_k ,

$$\xi(z_k + h) = \text{grad } w_c(y_k) + \text{Hess } w_c(y_k)DY_k(z_k)h + o(|h|).$$

Expanding the smooth function $(K_k)_y$ in its two variables therefore gives

$$\{\text{Hess } w_c(y_k) + (K_k)_{yy}\}DY_k + (K_k)_{yz} = 0.$$

Thus stationary differentiation uses the subgradient expansion even when neighboring endpoints are nonsmooth. Solving it with (5.32) gives, in parallel frames,

$$\text{Hess } w_c(y_k) = \text{Hess } G_k(z_k) + O(\tau_k). \quad (5.33)$$

In particular these endpoint Hessians are uniformly bounded. The error statement follows directly from the displayed short-cost expansions and the bounded inverse in (5.32); it does not assume a bound for $\text{Hess } w_c$ in advance.

Step 7: insert the true center form and take matrix limits. Set

$$r_k = \tau_k^{-1} \log_{y_k} z_k, \quad l_k = \varphi'_c(v(y_k)), \quad \hat{x}_k = \exp_{y_k} r_k, \quad \hat{p}_k = \log_{\hat{x}_k} y_k.$$

The true center endpoint is $\zeta_k = \exp_{y_k}(r_k/l_k)$, and it is before cut because $y_k \notin N$. Identify $T_{x_k}M$ with $T_{\hat{x}_k}M$ by parallel transport over their short joining segment, and let $B_k : T_{x_k}M \rightarrow T_{y_k}M$ be the exponential differential at $\hat{p}_k$ after that identification. Define

$$V_k = l_k B_k^* \{ \text{Hess } v(y_k) + A_{y_k}(r_k/l_k) \} B_k > 0. \quad (5.34)$$

The exact chain rule for the scalar modification gives

$$\begin{aligned} B_k^* \{ \text{Hess } w_c(y_k) + A_{y_k}(r_k) \} B_k &= V_k + B_k^* \{ A_{y_k}(r_k) - l_k A_{y_k}(r_k/l_k) \} B_k \\ &\quad + \frac{\varphi''_c(v(y_k))}{l_k^2} \hat{p}_k \hat{p}_k^*. \end{aligned} \quad (5.35)$$

All vectors on the last line are understood after the same isometric identification. Lemma 3.4 bounds the middle term below by $c(1 - l_k)\text{Id}$. Equation (5.33) and the vanishing pole mismatch show that the left side differs from H_k by $o(1)\text{Id}$. The verification is the same smooth coordinate comparison as in Lemma 5.4, now using the bounded unregularized Hessians in place of the smooth support Hessians. This proves

$$H_k \geq V_k + c \frac{b_k}{D} \text{Id} - C \frac{b_k}{D^2} p_k p_k^* - o(1)\text{Id}.$$

It also bounds V_k from above, because $V_k > 0$ and H_k is bounded.

Taking the determinant in (5.34) and using (5.4) gives the exact factorization

$$\det V_k = l_k^n \sigma_{\hat{x}_k}(\hat{p}_k)^2 \frac{|\det D(\exp. \text{grad } v)(y_k)|}{\sigma_{y_k}(r_k/l_k)} \geq c \frac{\sigma_{\hat{x}_k}(\hat{p}_k)^2}{\sigma_{y_k}(r_k/l_k)}. \quad (5.36)$$

The squared factor is the determinant of the Hessian pullback. All scalar derivatives lie between $1/2$ and 2 , so the constants in this density inequality are independent of K .

There is no assertion that the limiting endpoint is an Alexandrov point of v . We take limits of matrices at the good endpoints just selected. The center modified rays have a uniform extension at the fixed scales: $1 - l_k \geq cb_-/D > 0$, and the ray through $\hat{x}_k$ continues to the true endpoint through duration $1/l_k$. Compactness of this family of proper minimizing subsegments therefore gives

$$\sigma_{\hat{x}_k}(\hat{p}_k) \geq c_{\text{fixed}} > 0.$$

Since $0 < V_k \leq C_{\text{fixed}}\text{Id}$, equation (5.36) now implies

$$\sigma_{y_k}(r_k/l_k) \geq \frac{c \sigma_{\hat{x}_k}(\hat{p}_k)^2}{\det V_k} \geq c'_{\text{fixed}} > 0.$$

Continuity of the exponential differential excludes a conjugate limit of the true center rays. An ordinary cut limit may remain; the specified nonsingular branch suffices.

Finally take a subsequence so that $b_k, H_k, L_k, V_k, V_{1,k}$ and all configuration parameters converge, in a smooth orthonormal trivialization. Discard any additional zero limiting weights. The selected weight remains at least m_* , and its outward displacement retains the positive fixed-scale lower bound from Step 3. The barycenter identity then gives another surviving vertex, so the surviving difference span is still the same nonzero line. The selected modified and true outer branches were already uniformly nonsingular in Step 4; the center modified branch has its fixed-scale extension.

Equation (5.36) gives a positive lower determinant in the limit, so $V > 0$. Equations (5.6), (5.13), and (5.35) give (5.23).

At each selected outer endpoint in the diagonal sequence, formula (5.19), with $H_{o,k} = m_{1,k}L_k$, has a positive divided-action term. Its scalar curvature coefficient satisfies

$$-\frac{\varphi_o''(v(y_{1,k}))}{l_{1,k}^2} = -\frac{b_k B_o''((v(y_{1,k}) - a)/D)}{D^2 l_{1,k}^2} \geq cK b_k / D^2.$$

Passing this inequality and (5.20) along $V_{1,k} \rightarrow V_i$ gives (5.24) and the other inequality in (5.25). No lower Taylor jet is asserted at the limiting endpoint. The weights, directions, and section-level estimates pass through the same chosen subsequence and give (5.21)–(5.22). Their constants and the cap were fixed before K . Only δ, M , the rate of the prefix limit, and the Jensen radii depended on the fixed outer template. $\square$

5.6 The determinant gain

Lemma 5.9 (Oblique projection and the missing direction). *Suppose the matrices and vectors satisfy (5.21)–(5.24). Then*

$$\det V_i \geq cK \det V, \quad (5.37)$$

where $c > 0$ depends on the fixed center-template constants, the dimension, and m_* , and is independent of K .

Proof. Put $\beta = Cb/D^2$, increasing C if necessary, and $\gamma = cKb/D^2$. Define

$$P = \text{Id} - ep^*/q.$$

This is the projection onto $p^\perp$ along e . Since $LE = 0$, $P^*LP = L$. On $p^\perp$ the center inequality gives $H \geq V$, and $L \geq H$. Thus

$$V_i \geq m_i P^* V P + \gamma p_i p_i^*. \quad (5.38)$$

Also $H(e, e) \leq L(e, e) = 0$ implies $V(e, e) \leq \beta q^2$.

Let Q have orthonormal columns spanning $p^\perp$, and use the nonsingular basis matrix $[Q, e]$. In this basis write

$$[Q, e]^* V [Q, e] = \begin{pmatrix} A & z \\ z^* & h \end{pmatrix}, \quad S = h - z^* A^{-1} z > 0.$$

Then $S \leq h = V(e, e) \leq \beta q^2$. The determinant of the right side of (5.38), in the same basis, is $m_i^{n-1} \det A \gamma (p_i \cdot e)^2$. Taking the ratio cancels the determinant of the nonorthogonal change of basis and gives

$$\frac{\det V_i}{\det V} \geq m_i^{n-1} \frac{\gamma (p_i \cdot e)^2}{S} \geq m_i^{n-1} \frac{\gamma}{\beta} \frac{(p_i \cdot e)^2}{q^2}.$$

Since $p_i = p + d_i e$ with $d_i > 0$ and $q > 0$, we have $p_i \cdot e \geq q$. Now use $m_i \geq m_*$ and $\gamma/\beta \geq cK$. $\square$

Proposition 5.10 (Contradiction to a non-power oscillation). *The section-oscillation function of Lemma 4.2 has a positive power upper bound at zero.*

Proof. Suppose otherwise, and take the sequence of scales supplied by that Lemma. Fix all center and weight constants, then choose K and its outer template, and then take sufficiently small scales to apply Proposition 5.8. Lemma 5.9 and (5.25) imply

$$cK \frac{\sigma_x(p)^2}{\sigma_y(r/l)} \leq \det V_i \leq C \frac{\sigma_x(p_i)^2}{\sigma_{y_i}(r_i/l_i)}.$$

The radial comparison of Corollary 3.3 gives

$$\sigma_y(r/l) \leq C\sigma_x(p), \quad \sigma_{y_i}(r_i/l_i) \geq c\sigma_x(p_i).$$

For the selected outward endpoint, (5.21)–(5.22) give $m_i \geq m_*$ and $q^2 \geq cD$. Its norm increment satisfies $|p_i|^2 - |p|^2 \leq 3D$ by Lemma 4.9. Corollary 3.6 therefore applies with $\mu = m_*$, $\kappa = c$, and $L_0 = 3$, and gives $\sigma_x(p) \geq c'\sigma_x(p_i)$. These parameters were fixed before K . Hence

$$cK\sigma_x(p) \leq C\sigma_x(p_i) \leq C'\sigma_x(p).$$

The center modified branch is nonconjugate, so $\sigma_x(p) > 0$. This bounds K by a constant fixed before it was chosen. Choosing K larger gives a contradiction. $\square$

6 Completion of the uniform estimate

Proposition 5.10 was applied to the section-oscillation supremum over the entire class of densities with the fixed bounds λ, Λ on (M, g) . It therefore gives constants $C_0 < \infty$, $\beta > 0$, and $r_0 > 0$, common to that class, such that

$$F(r) \leq C_0 r^\beta \quad (0 < r < r_0).$$

Lemma 4.1 now gives a common Hölder estimate with $\alpha = \frac{1}{2} \min\{\beta, 1\}$. In particular, every pole has a single contact endpoint. Applying the same estimate to the reversed density pair gives singleton contacts on the other side, with the same constants. The two contact maps are continuous inverses and agree almost everywhere with the optimal maps. Enlarging the Hölder constant to cover distances outside the small-scale range proves Theorem 1.1.

Remark 6.1. The proof produces a positive exponent by contradiction and compactness. It gives no explicit exponent or estimate uniform over varying metrics or density bounds. The inverse map comes from applying the result to the reversed positive-density pair; no time-one injectivity is asserted for an arbitrary cost potential.

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